計算認知神經科學
大腦神經科學

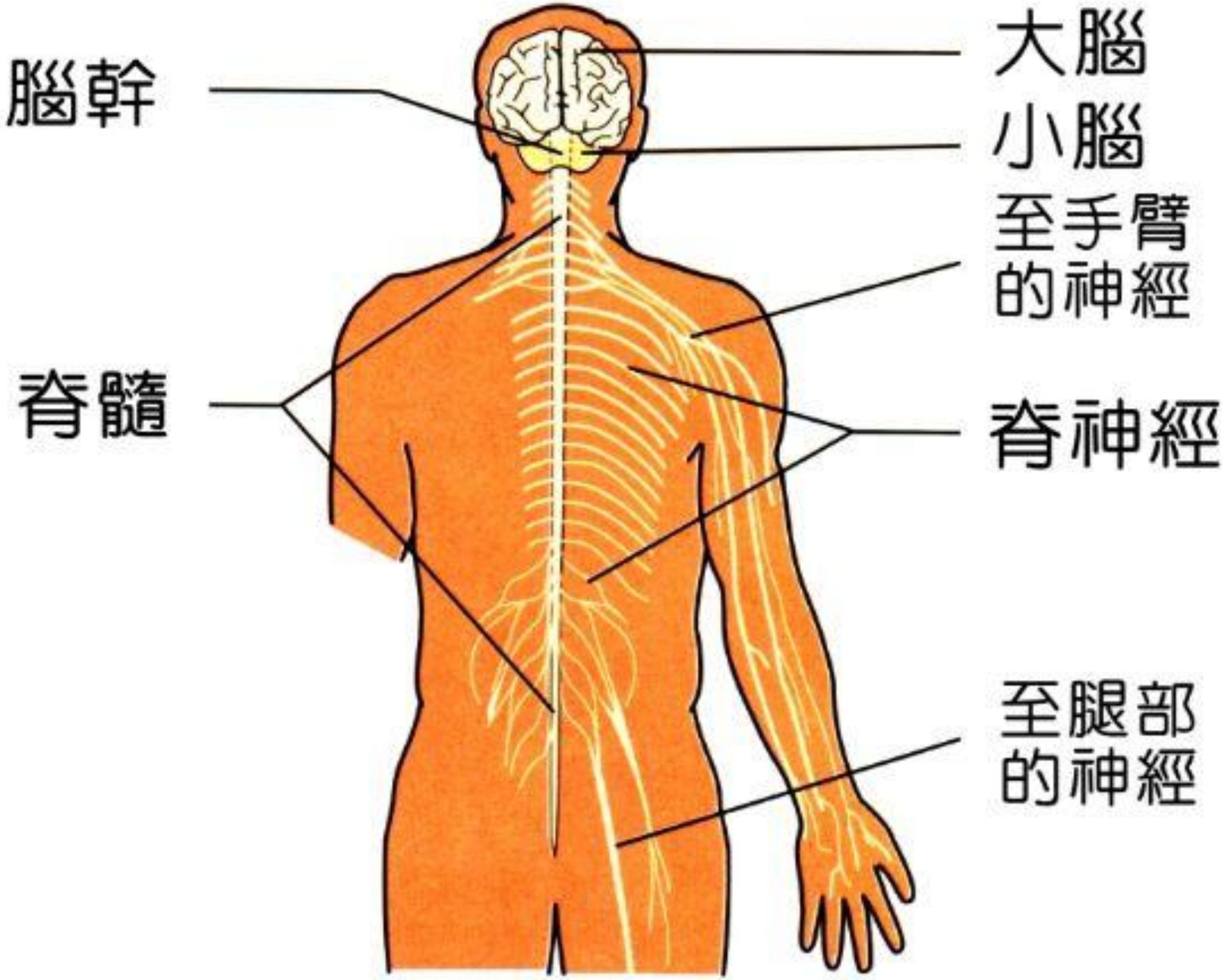


圖5-2 人體的神經系統

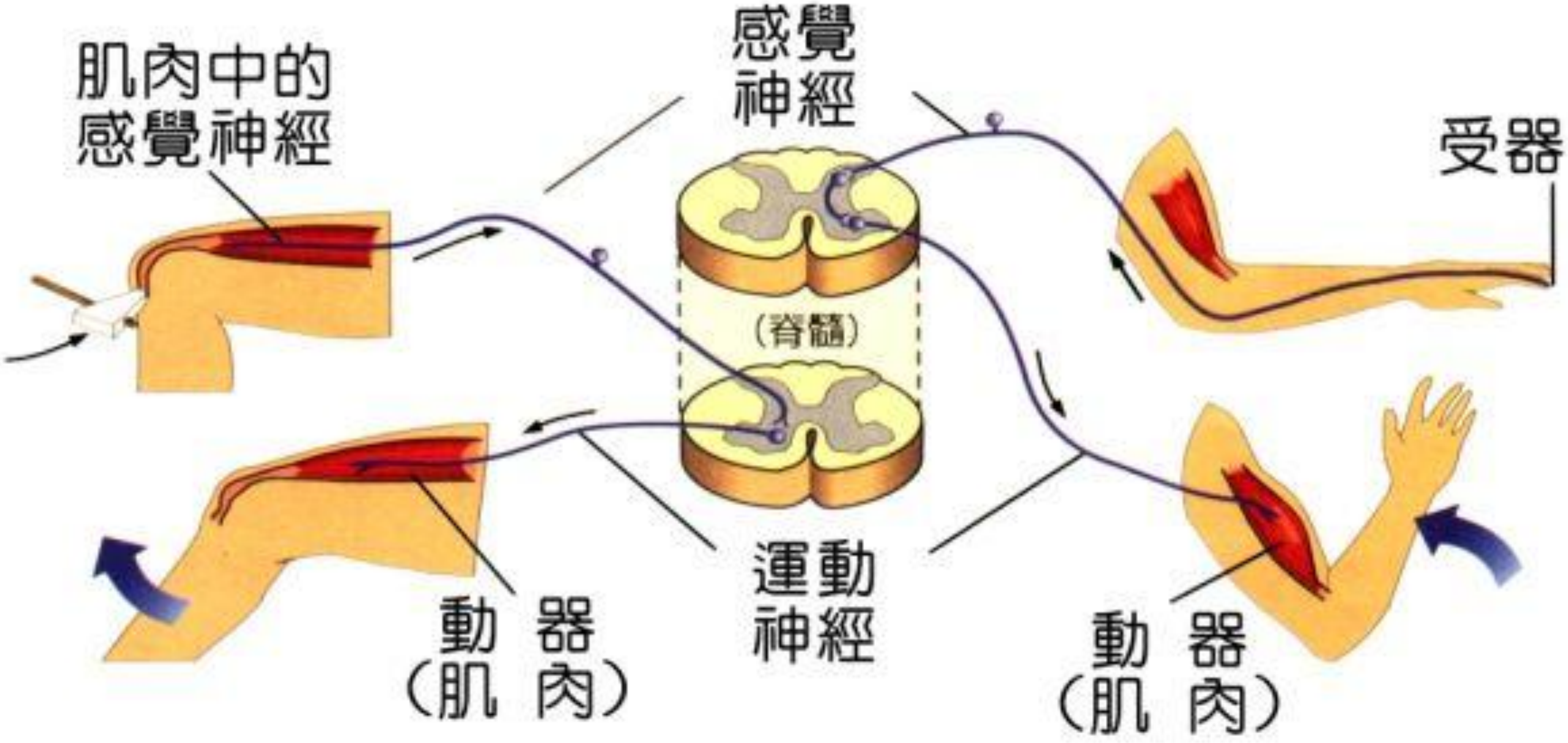


圖5-4 脊髓橫切面和脊反射

中樞神經系統(腦與脊髓)

大腦

小腦

間腦

腦幹

（延髓、橋腦與中腦）

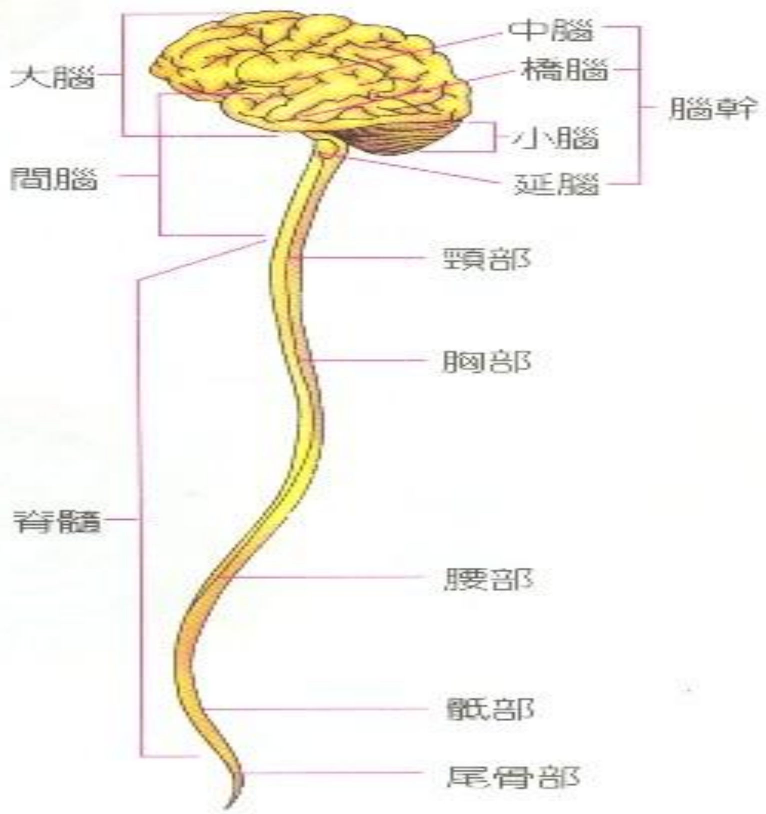
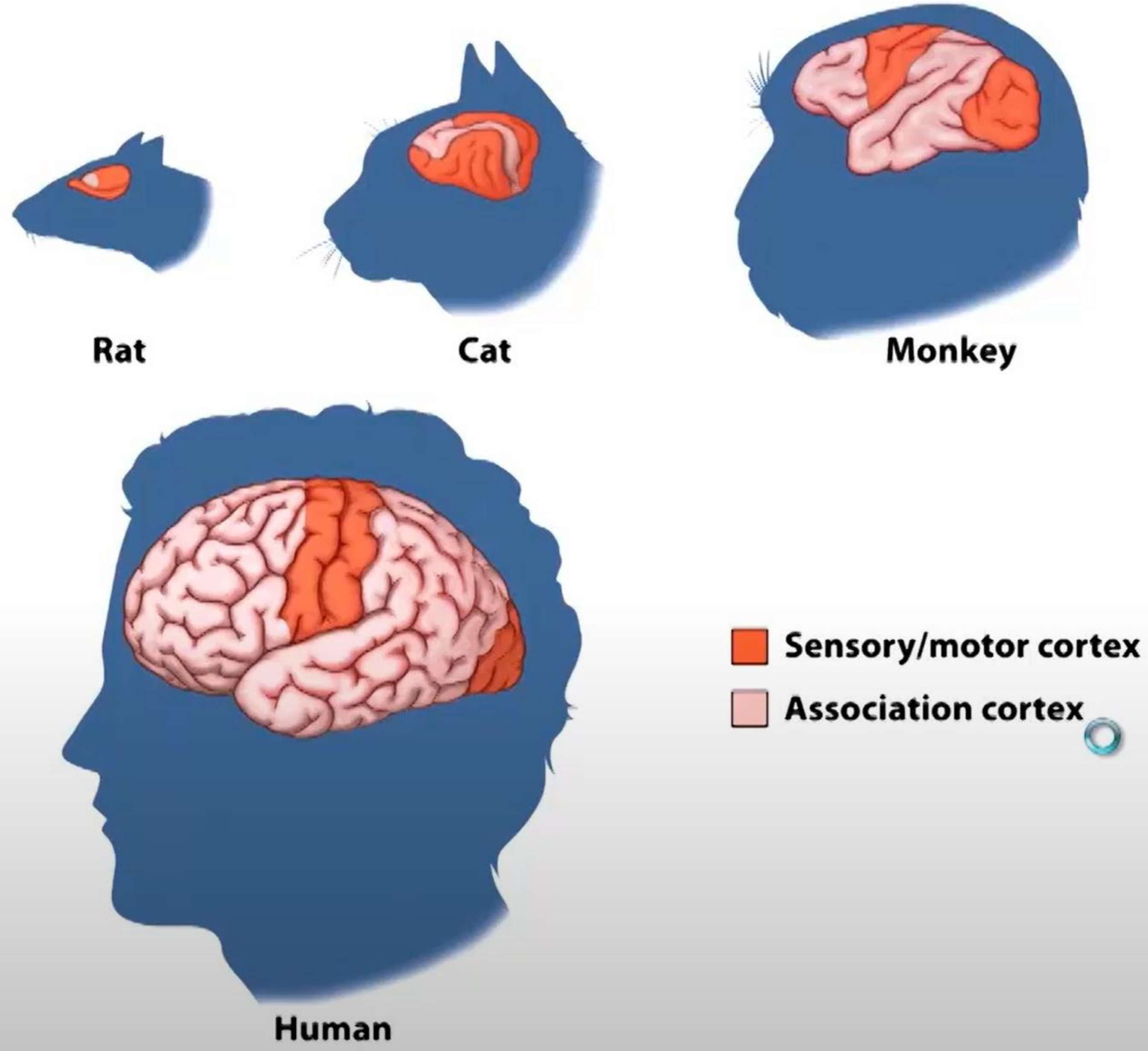


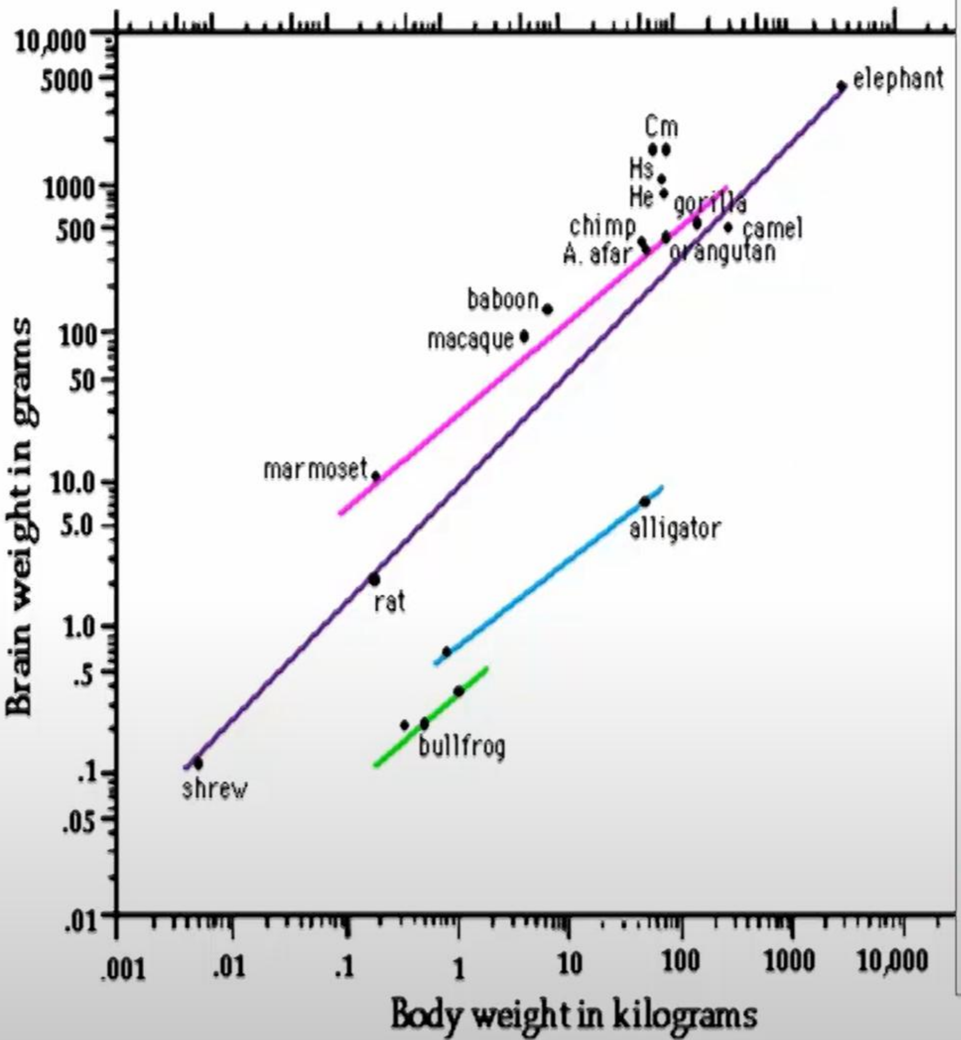
圖 6-2 腦及脊髓之結構圖

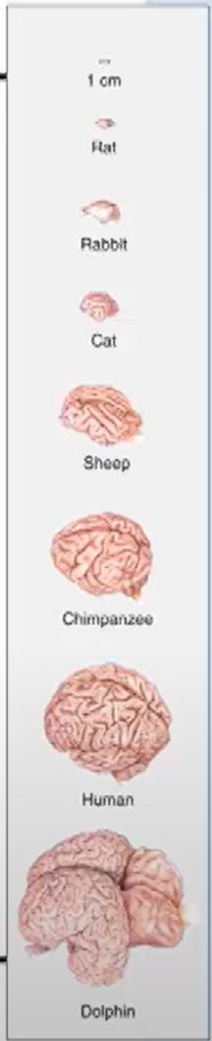
Mammalian evolutionary development

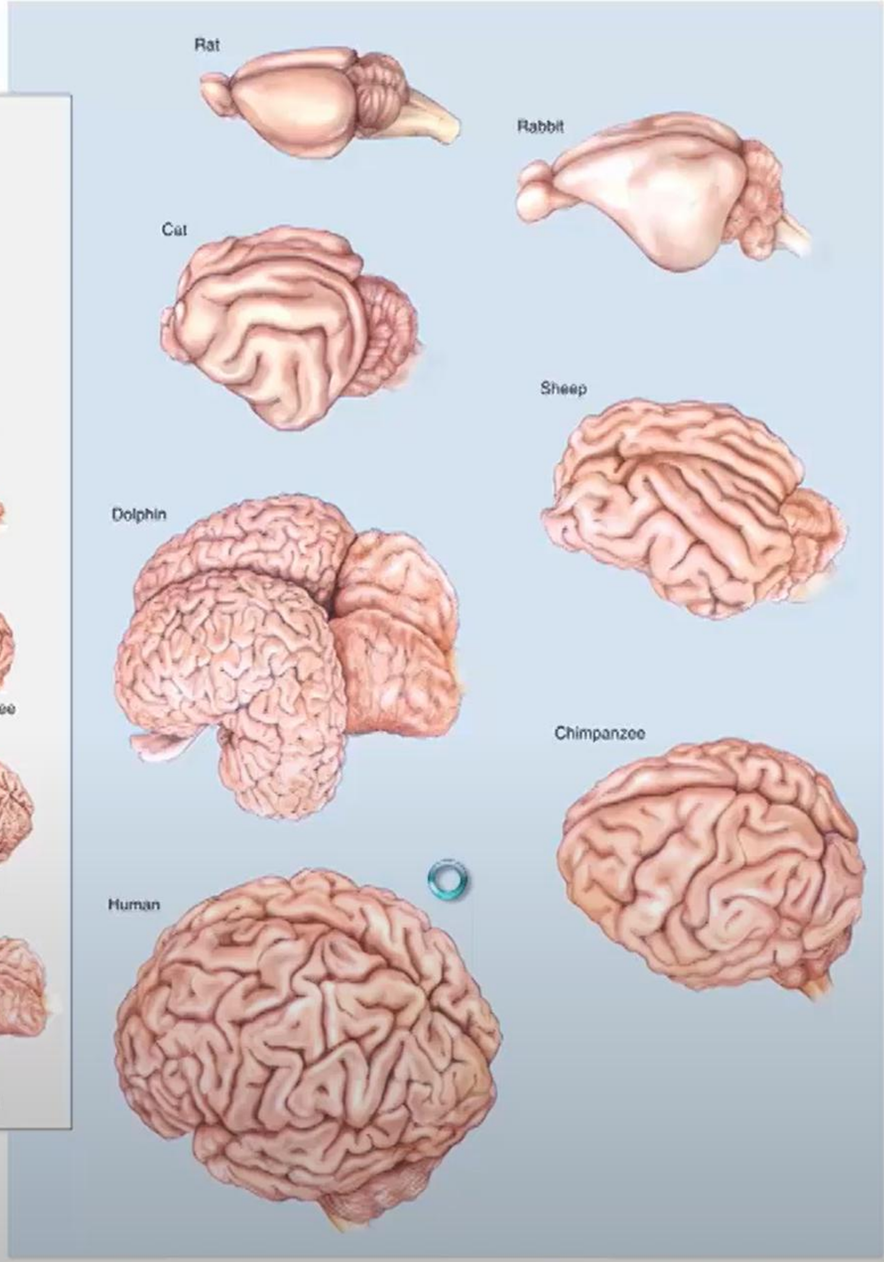


Mammalian evolutionary development

頭大腦大就是聰明?





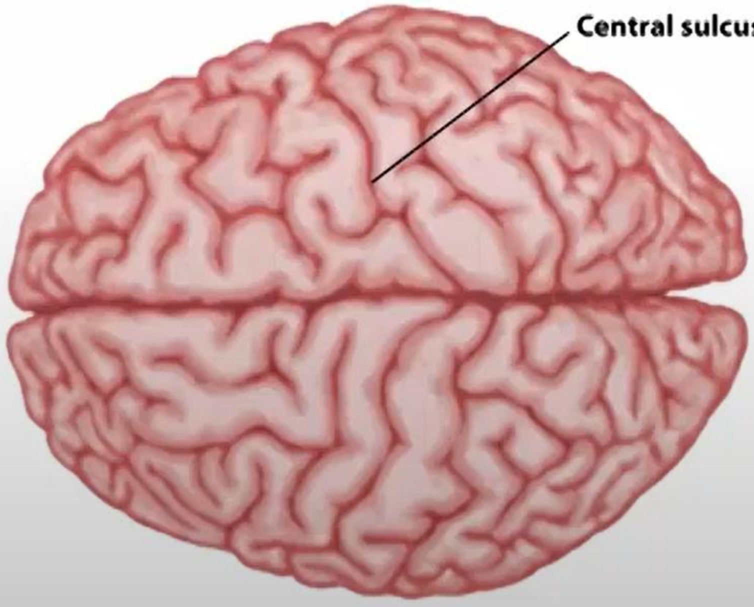


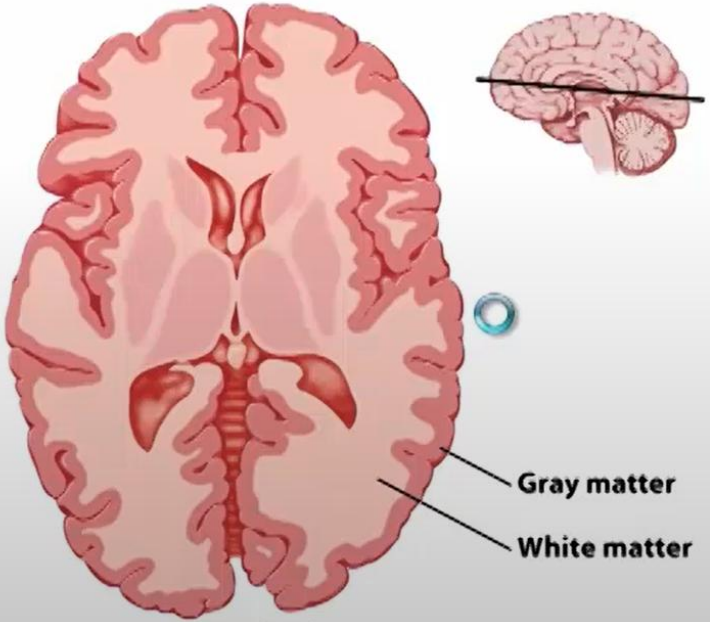
取自陽交大生物科技學系 黃植懋老師
取自台大心理系 黃從仁教授

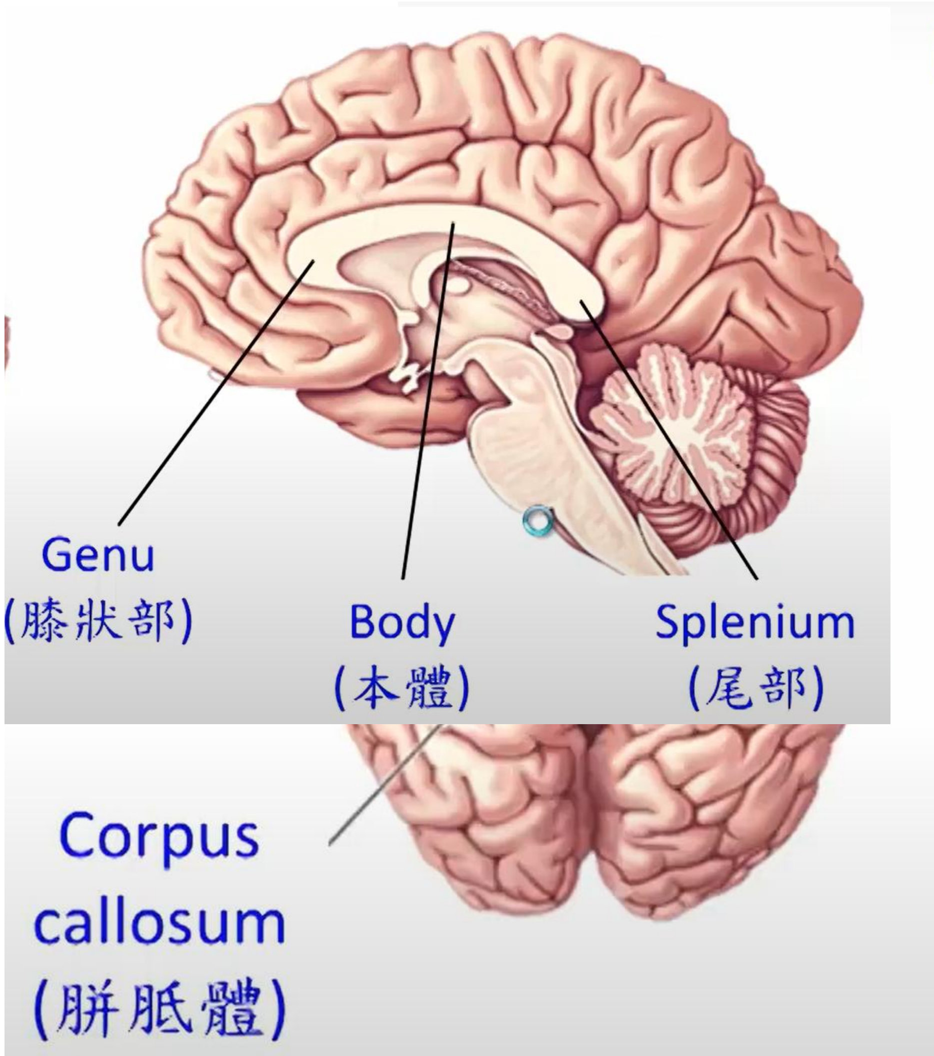
取自 國立清華大學生命科學系潘榮隆榮譽講座教授

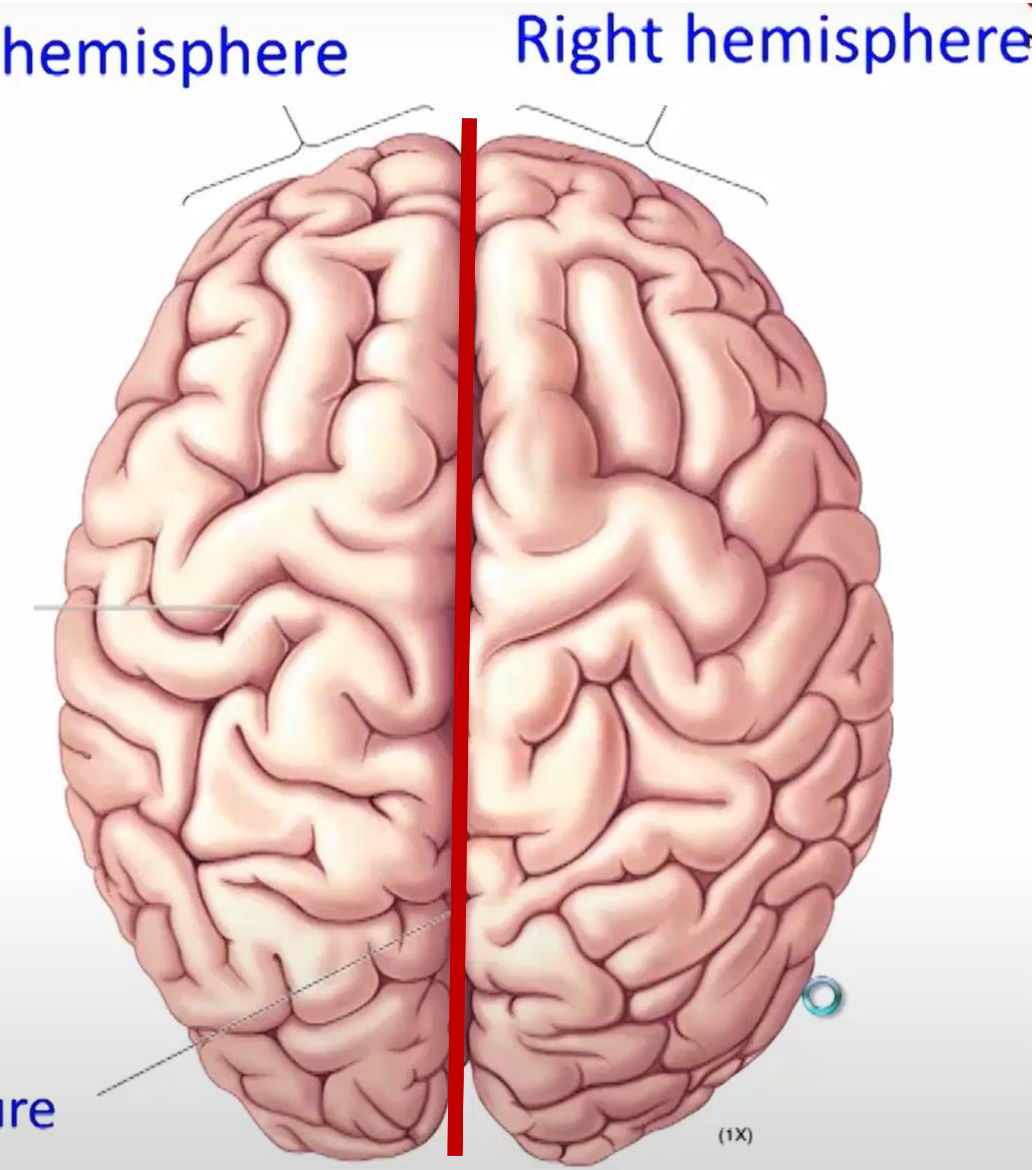
Cerebral cortex

- 2 hemispheres
- 3mm sheets of neurons sitting atop core structures
- Many folds – gyri and sulci 腦迴與腦溝
- Gray matter (neuronal cell bodies) and white matter (axons)









Longitudinal fissure: divides the two cerebral hemispheres

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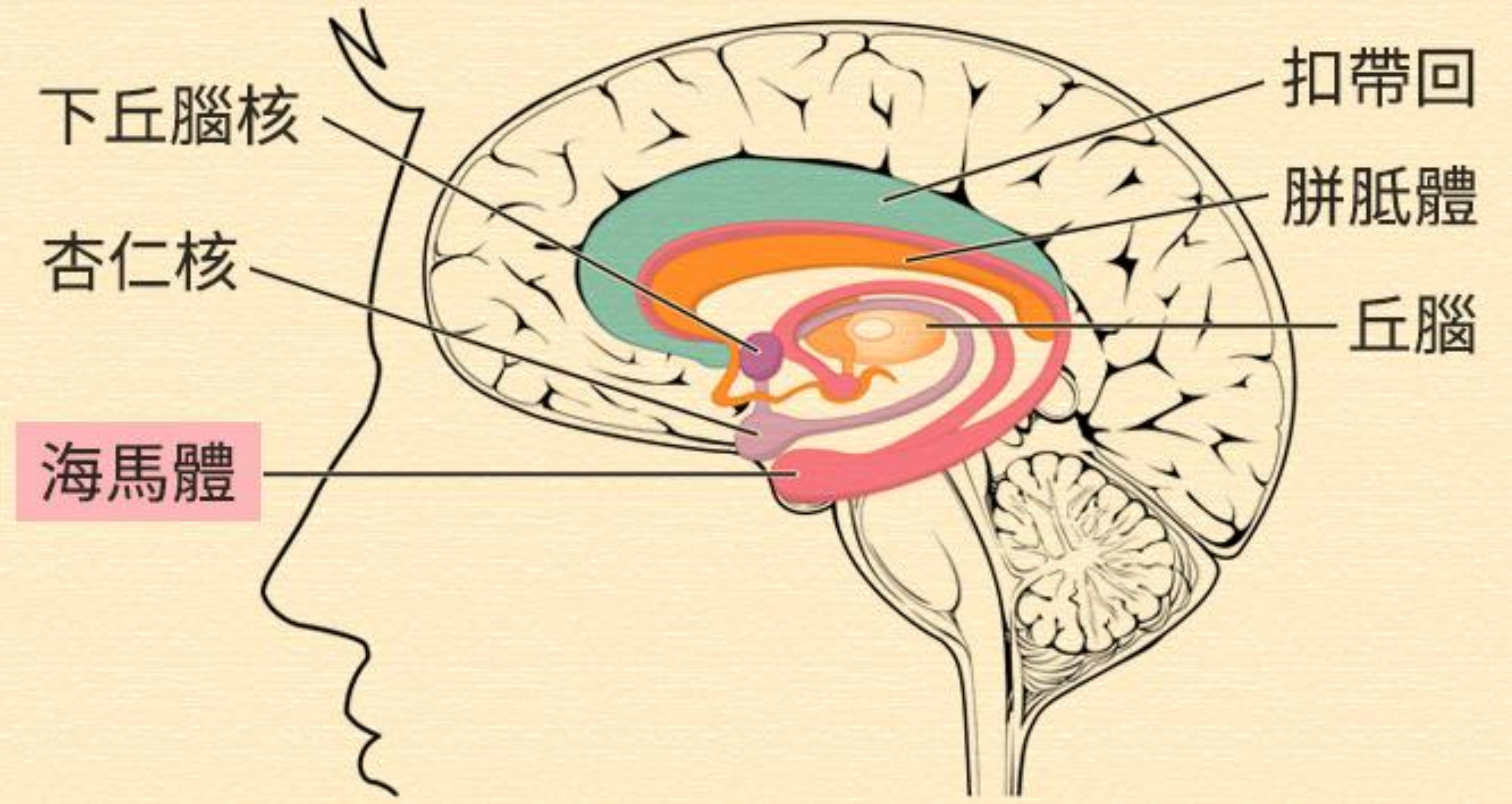
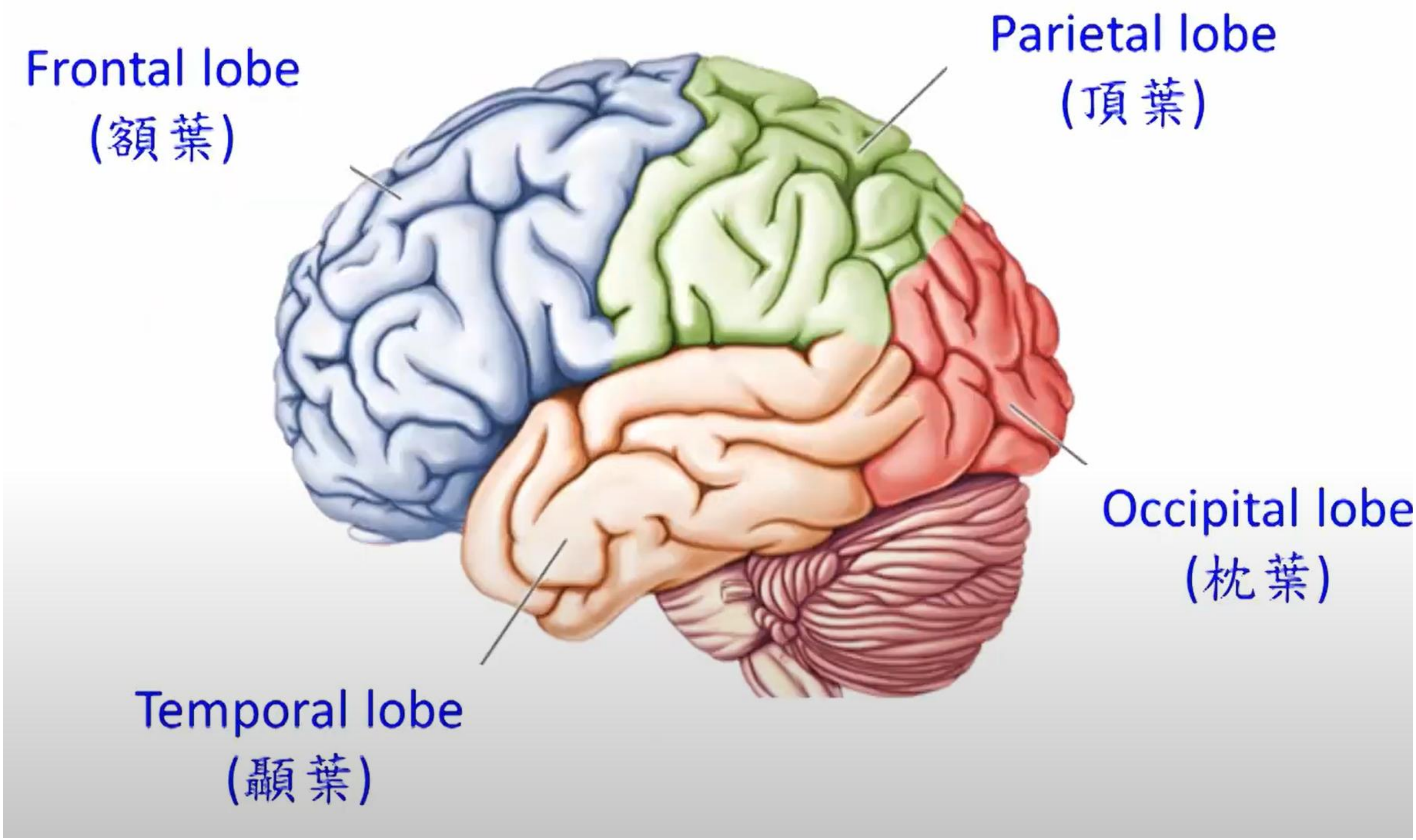
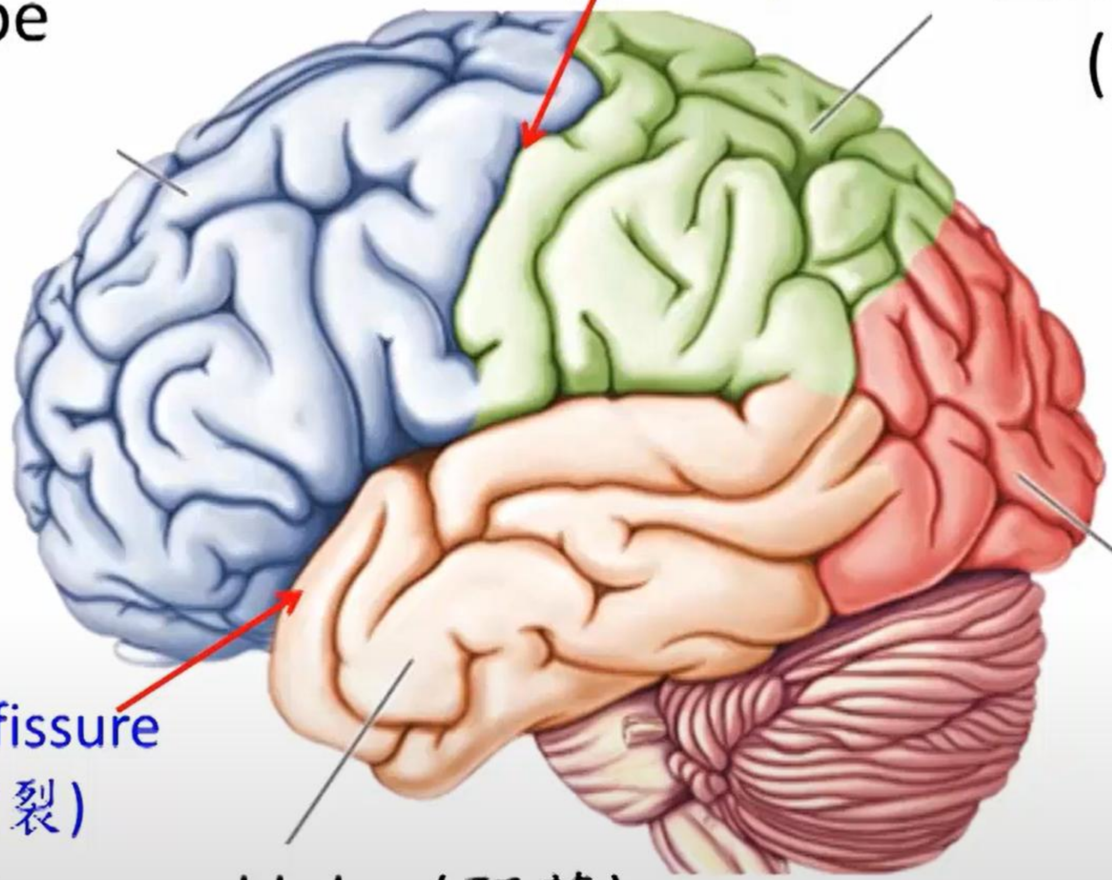
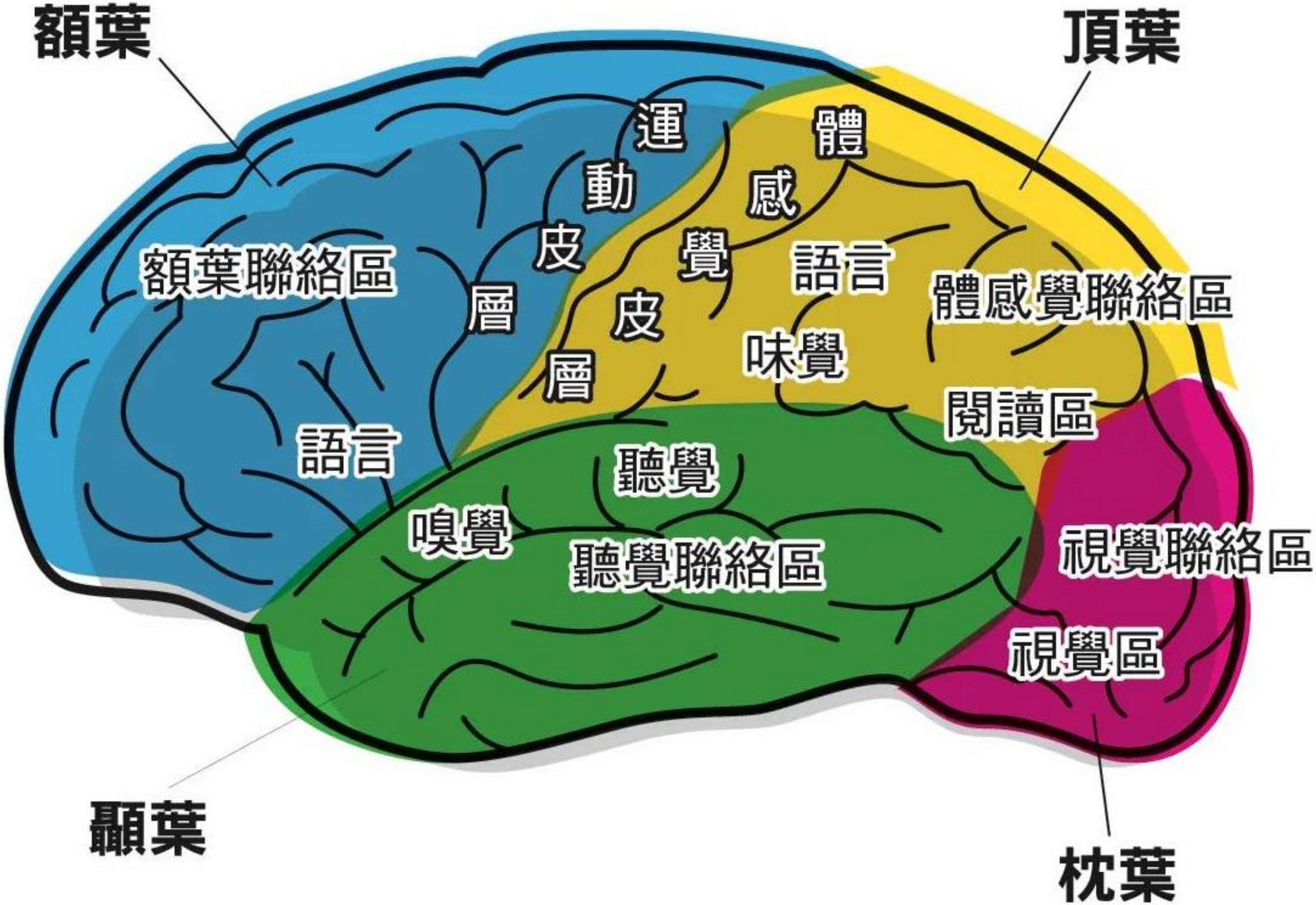

下丘腦核
杏仁核
海馬體
扣帶回
胼胝體
丘腦


Frontal lobe
(額葉)
Parietal lobe
(頂葉)
Occipital lobe
(枕葉)
Temporal lobe
(顳葉)

Four lobes


Central sulcus
(中央溝)
Parietal lobe
(頂葉)
Frontal lobe
(額葉)
Occipital lobe
(枕葉)
Lateral (Sylvian) fissure
(外側裂/腦側裂)
Temporal lobe (顳葉)



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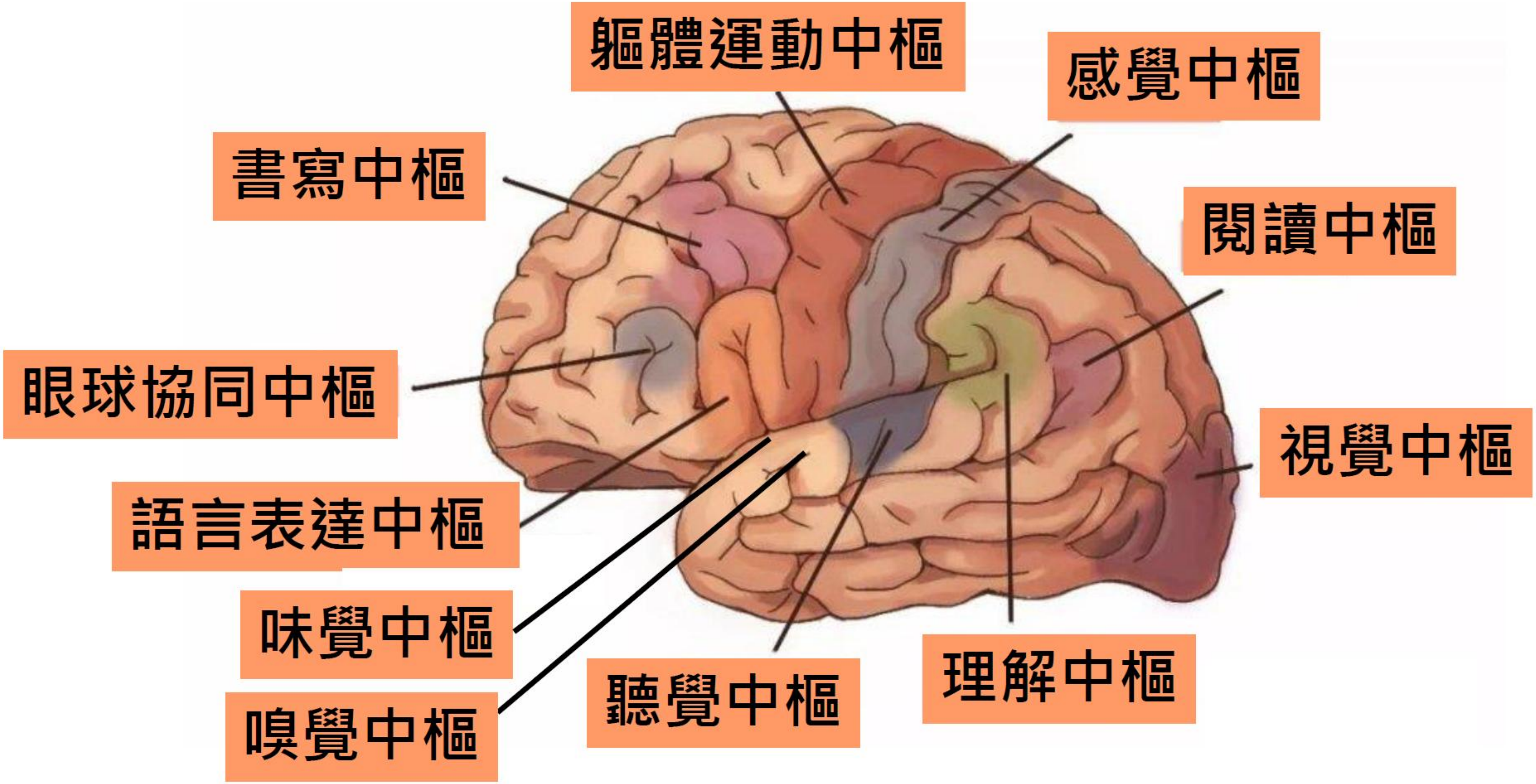
取自 國立清華大學生命科學系潘榮隆榮譽講座教授

人的大腦皮質功能區圖

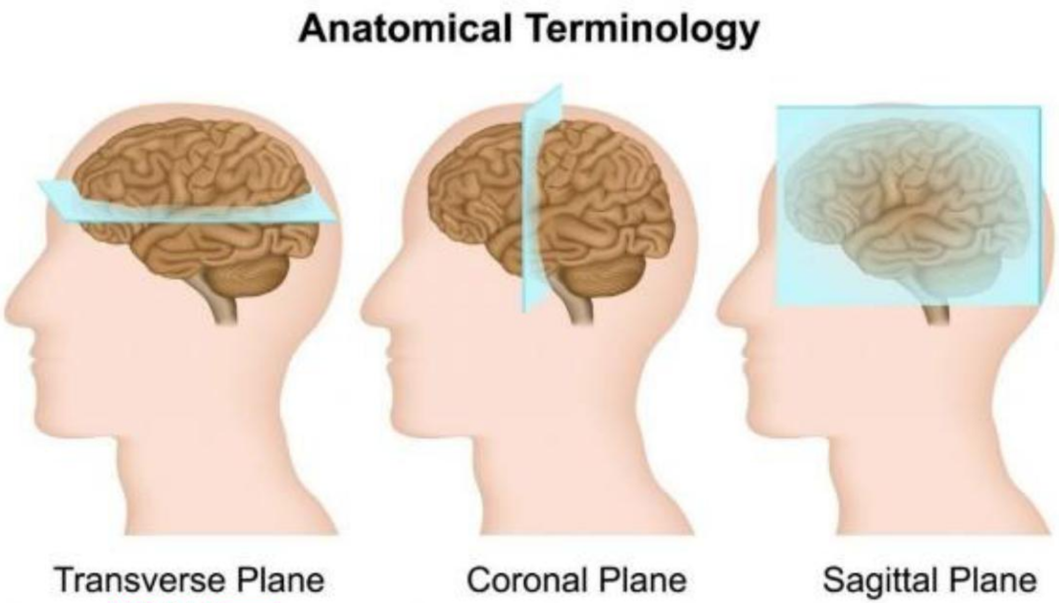


(https://read01.com/Q3RALDM.html)

Cells of the nervous system

- 2 classes of cells in nervous system
  - Neural cells (Neurons)
    - ✓ Specialized cells capable of transmitting information over long distances
    - ✓ Considered to be the functional cells of the brain
    - ✓ Typically do not reproduce: but not always the case
  - Glial Cells
    - ✓ Structural support
    - ✓ Insulation to neurons to make neural transmission more efficient

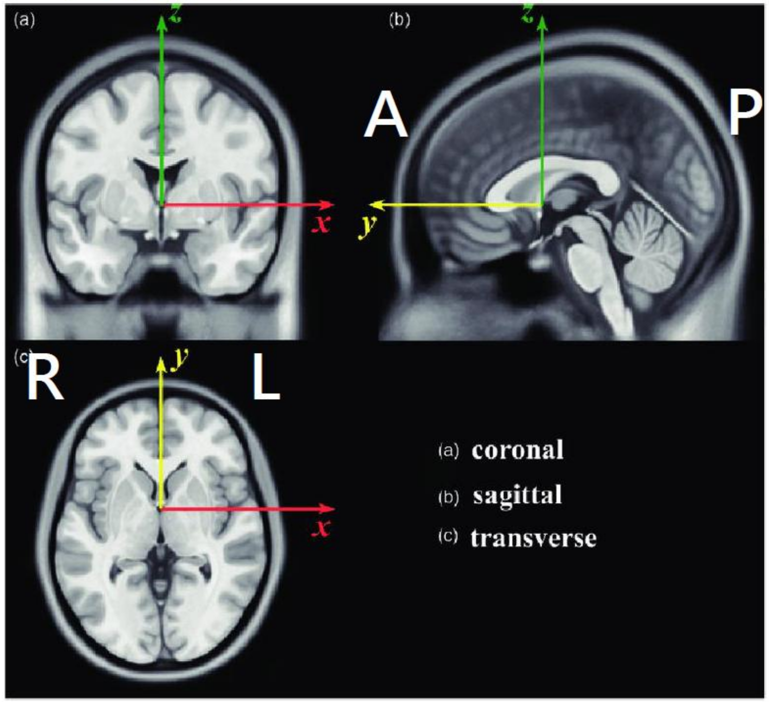
腦部剖面圖
（MNI Model）



橫斷面 冠狀面 矢狀面

（https://reurl.cc/g4aQyb）

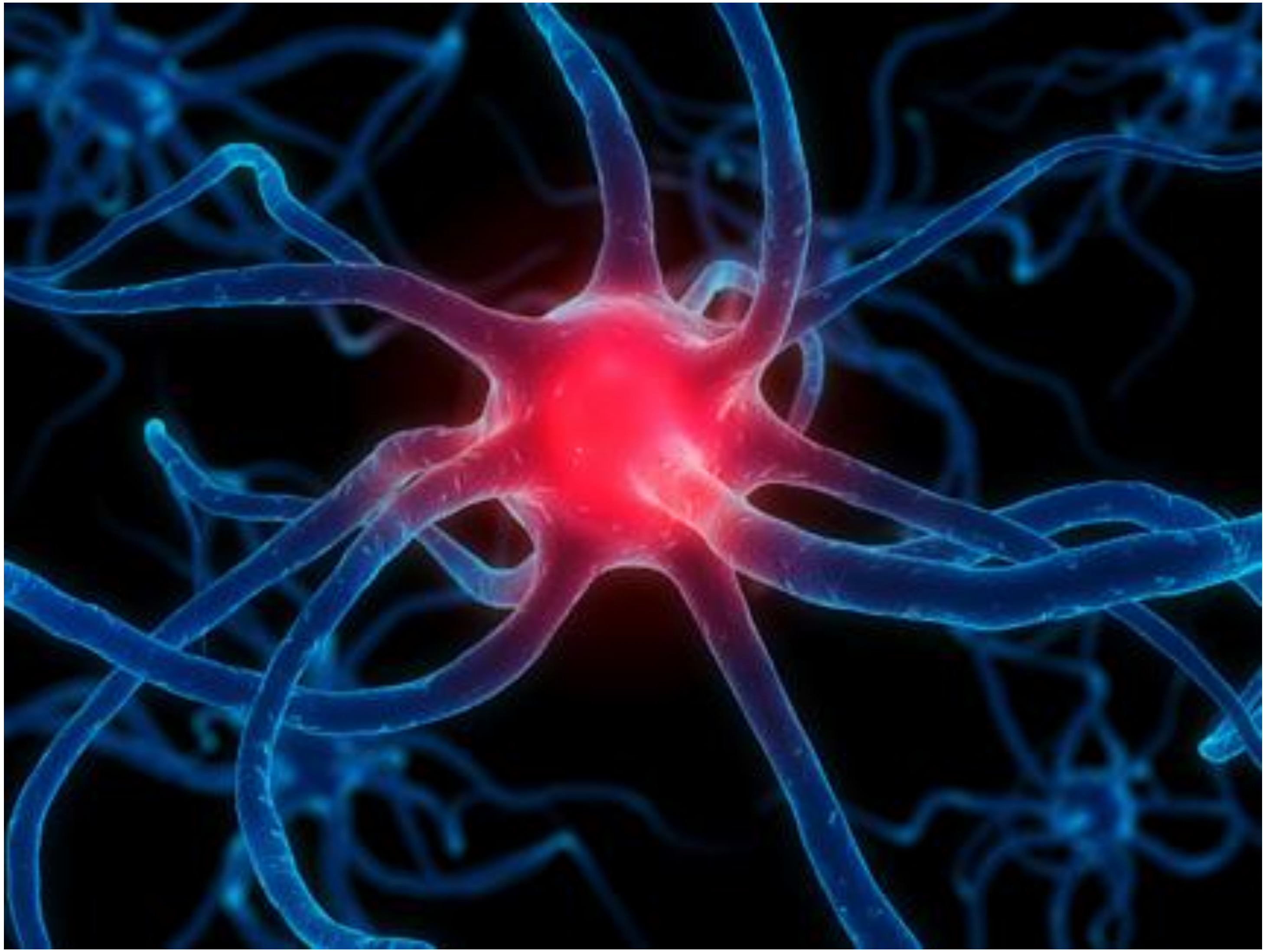
冠狀面 矢狀面



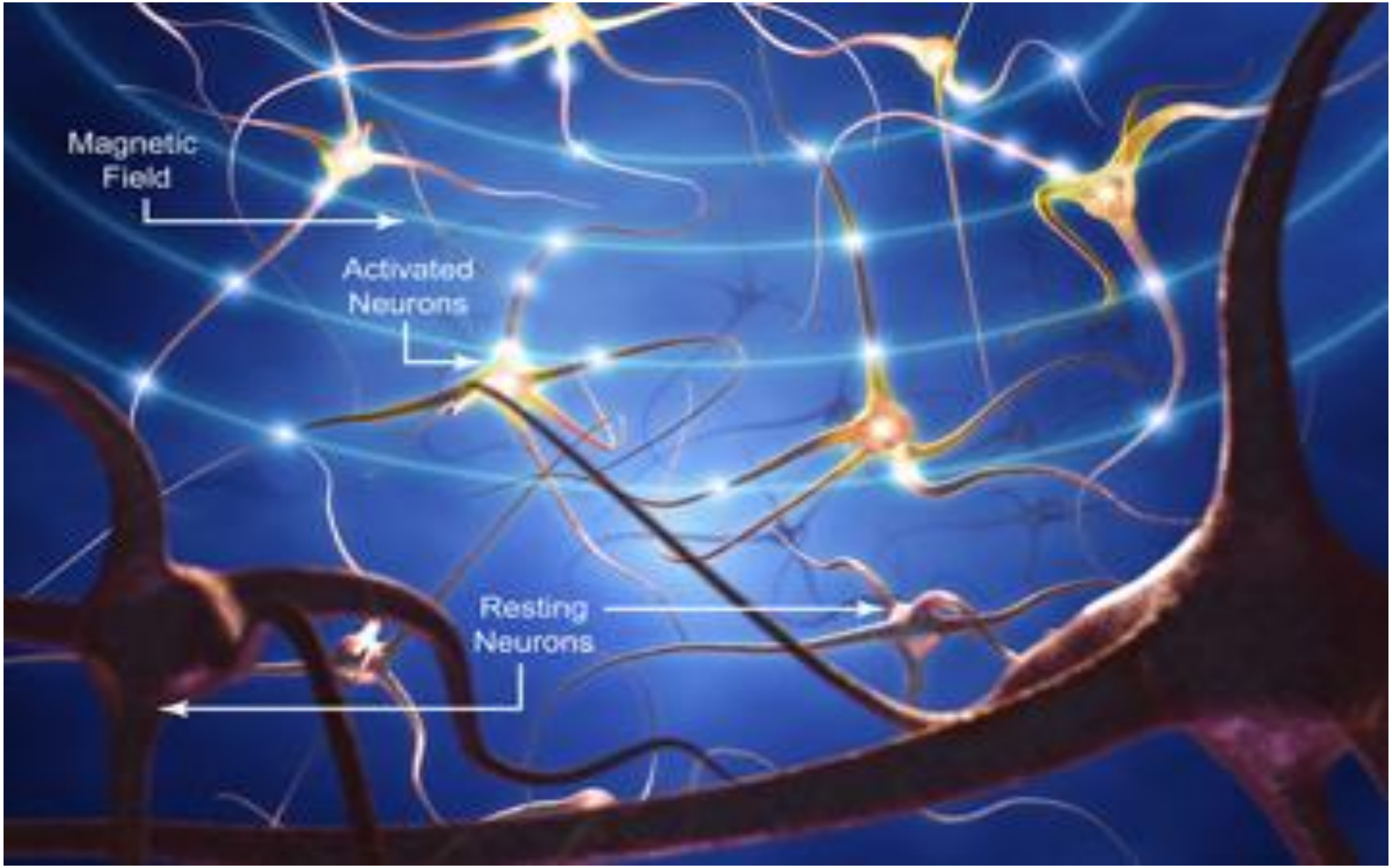
橫斷面

（https://www.researchgate.net/figure/The-MNI-coordinate-system-of-the-brain_fig1_344782562）

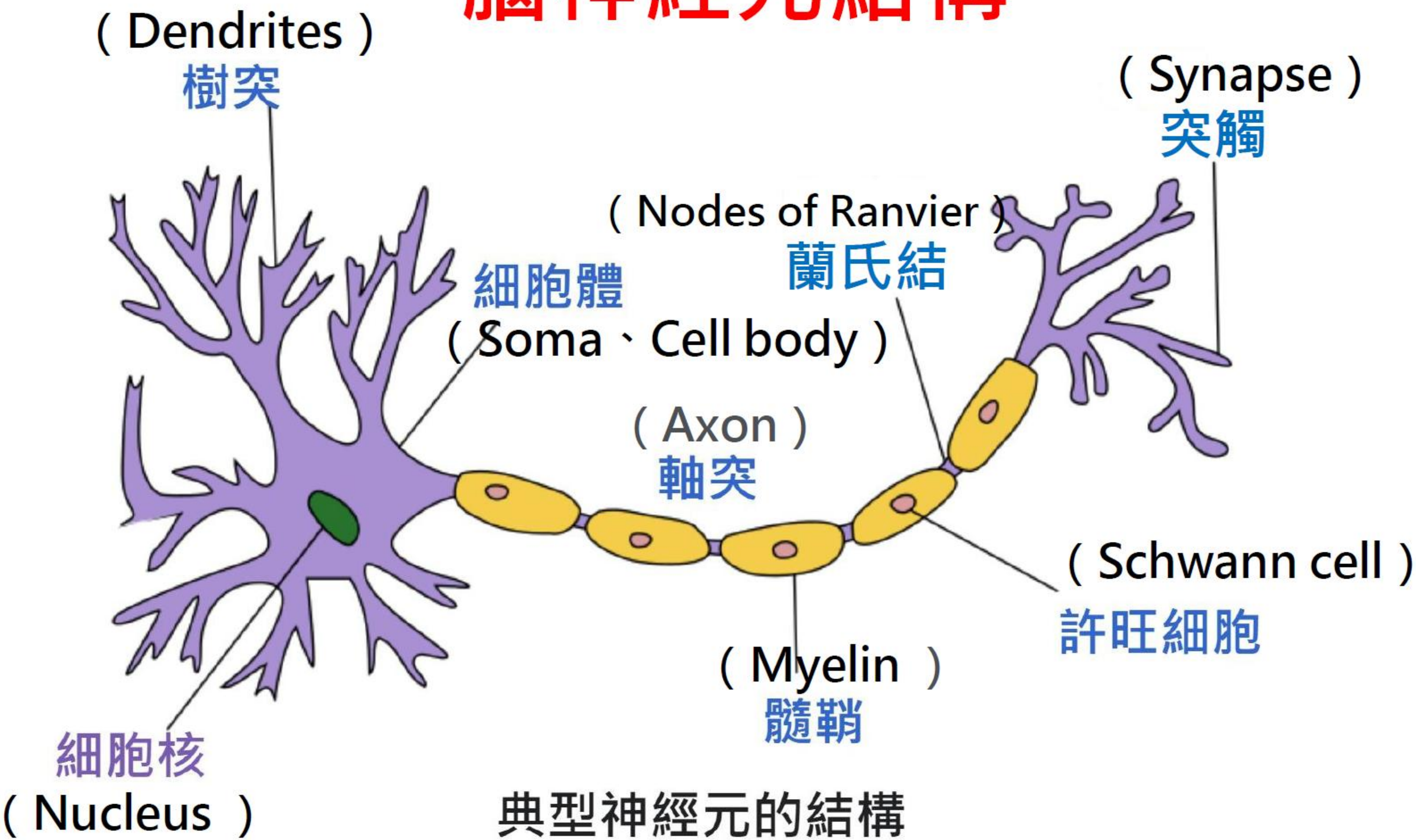
單個腦神經圖



幾個交接的腦神經



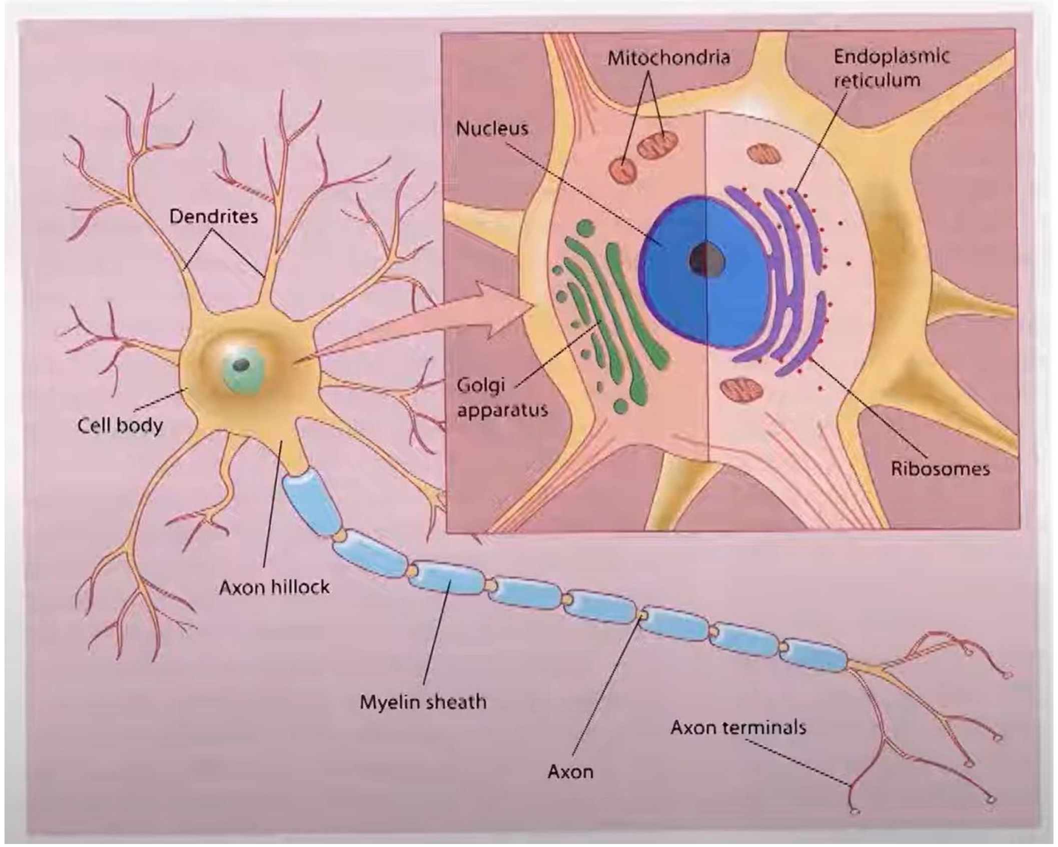
腦神經元結構



典型神經元的結構

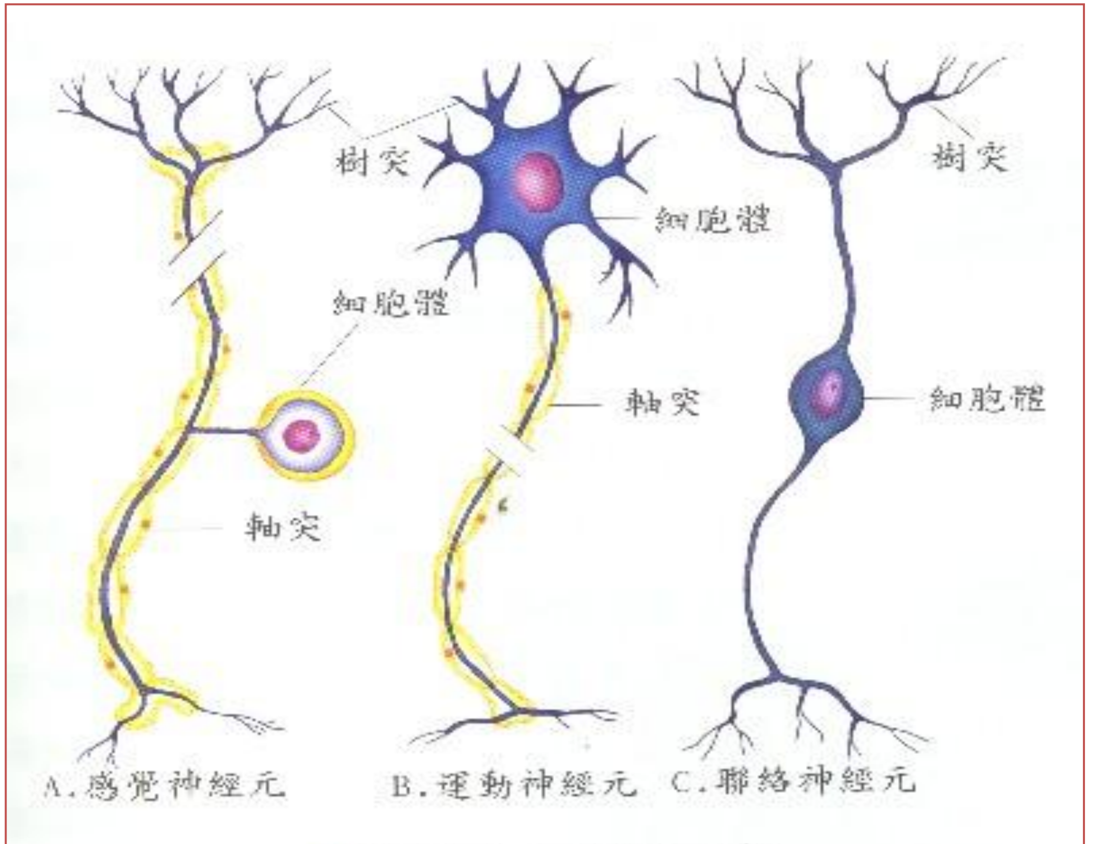
Neurons

- About 100 billion neurons in the brain (~ 10^{11})
- Distinguished by
  - ✓ Form (morphology)
  - ✓ Function
  - ✓ Location
  - ✓ Interconnectivity



神經元種類

感覺神經元
運動神經元
聯絡神經元
神經元之構造
（細胞體）
（樹突）
（軸突）



▲圖 6-10　神經元的形態

Glial cells

- Greek "glue"
- More numerous than neurons: 10^{12}
- Both in central (CNS) and peripheral (PNS) nervous system
- Although there are many neurons in the human brain (about 100 billion), glia outnumber neurons by tenfold.

10% of the human brain?

人類有多少腦細胞？

- ✓ 重要的不是腦細胞數目的多少，是腦細胞之間的連結數的多寡–每個腦細胞可以長出多達二萬條的「細樹枝」或「樹狀突」。
- ✓ 每一個腦細胞約與5000到50000個腦細胞有接觸。
- ✓ 據專家說我們的腦子可容納大約10億冊1000頁的書。
- ✓ 腦為身體老化最快的部分。腦細胞約有150～160億個，但過了20歲以後，平均一天會死10萬個腦細胞，並沒有新的腦細胞來補充。
- ✓ 老年人衹剩3／4以下的腦組織。但是腦細胞如果經常利用，譬如看書、寫文章、思考等等，可以延長腦細胞的生命（其他肌肉、骨骼細胞若經常活動，也能延長細胞減少的期限）。

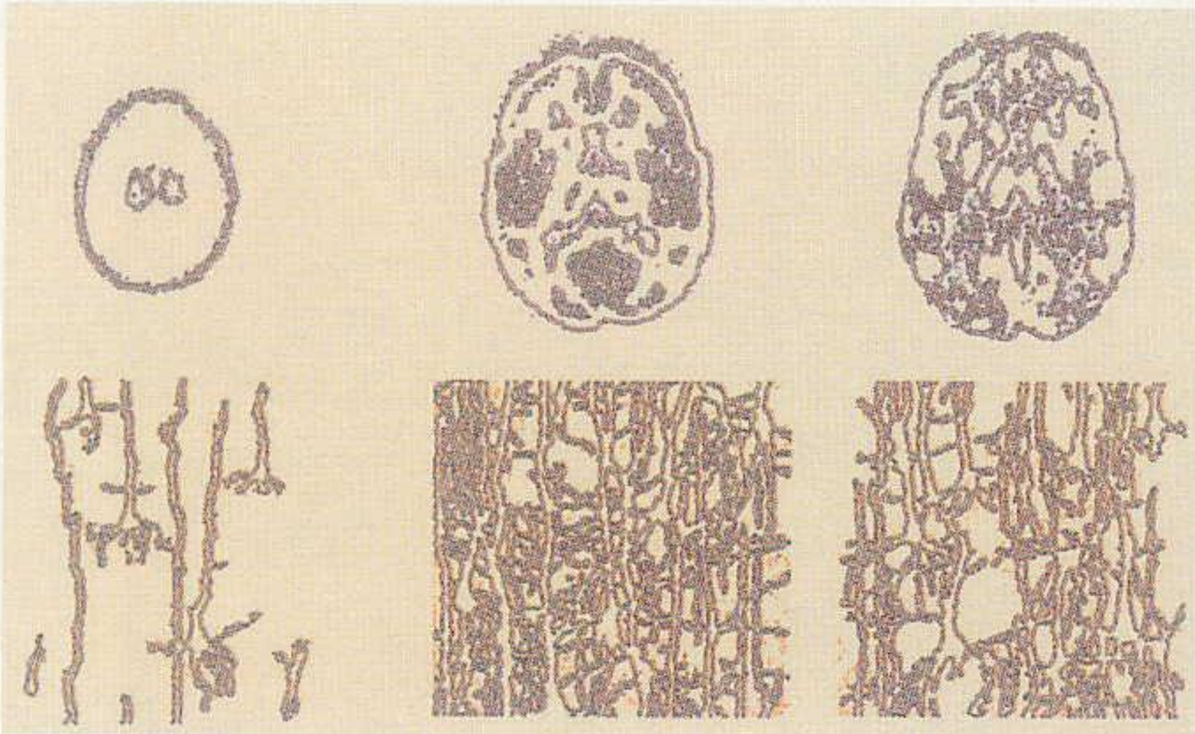
Age-related changes in neural structure

- Fluorescent cell staining

Visual cortex: Kitty

Visual cortex: old cat

腦神經連結



神經連接在出生時很稀疏（左圖），嬰兒期長得非常快，到六歲大（中圖）已經達到最大密度。此後神經密度開始減弱，因為不需要的連接開始死亡（右圖）。成人只要繼續使用大腦、學習新的東西，神經連接數量就會增加。假如腦部沒有持續使用，連接會更少。

摘自：大腦的秘密檔案 洪蘭 譯

取自陽交大生物科技學系 黃植懋老師
取自台大心理系 黃從仁教授

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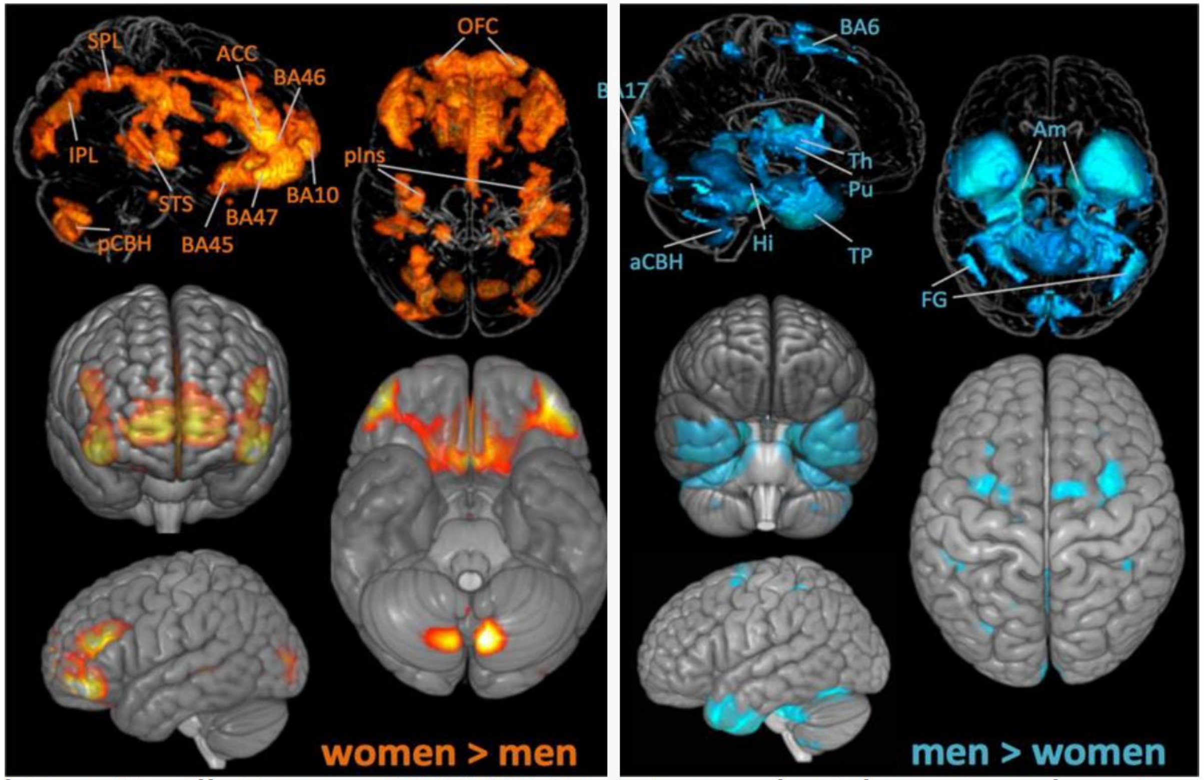
男女大腦有別



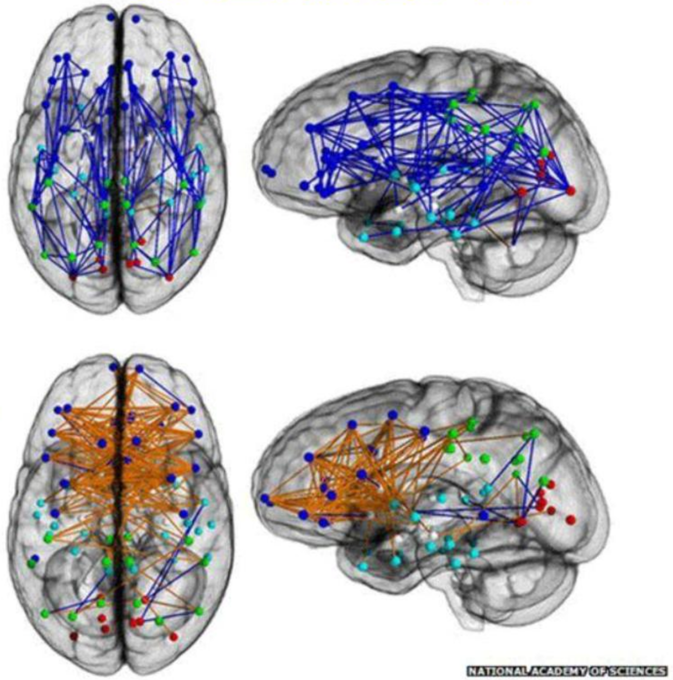
(https://quillette.com/2019/03/11/science-denial-wont-end-sexism/)

【腦科學揭露女人思考的秘密：洪蘭 Daisy L. Hung @TEDxTaipei 2015】
https://youtu.be/wWnUczsfGv0

「神經網路圖」

男性

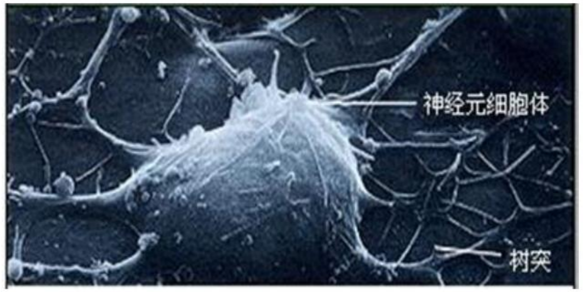
女性



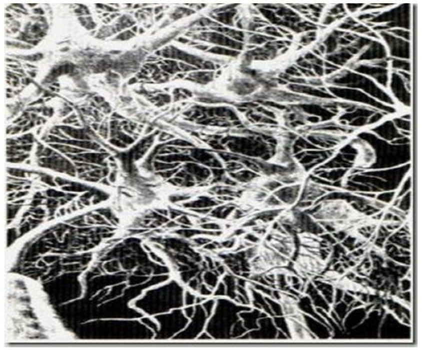
男性腦神經連接走向由前至後，
左右腦間沒什麼連接，
女性腦神經左右腦之間迂迴曲折地連接。

大腦神經的連結

860 - 1000 億個神經元 / 大腦



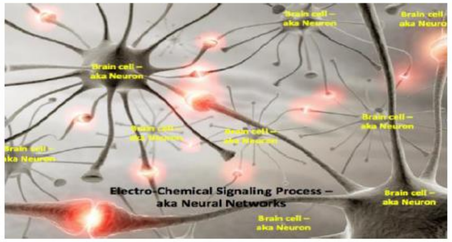
每個神經元與 > 5 萬個連結
共 > > 4000萬億個連結



每個連結有 on / off
共計　至少有

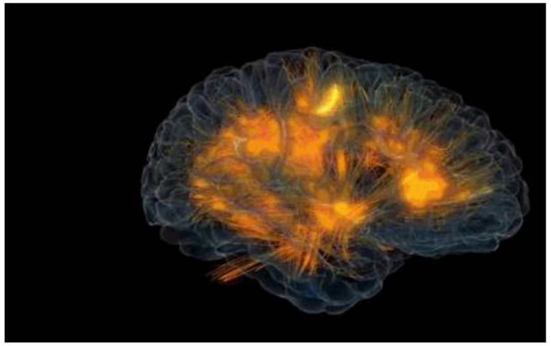
$> > 2^{4000\text{萬億}}$

(Source： Radu Jianu / Brown University)



小小觸動

蝴蝶效應



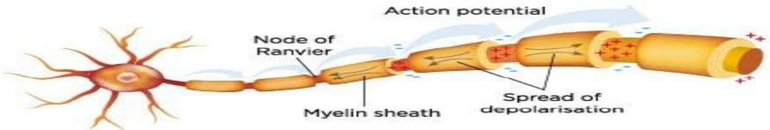
(https://tw.aboluowang.com/2020/1024/1515616.html)

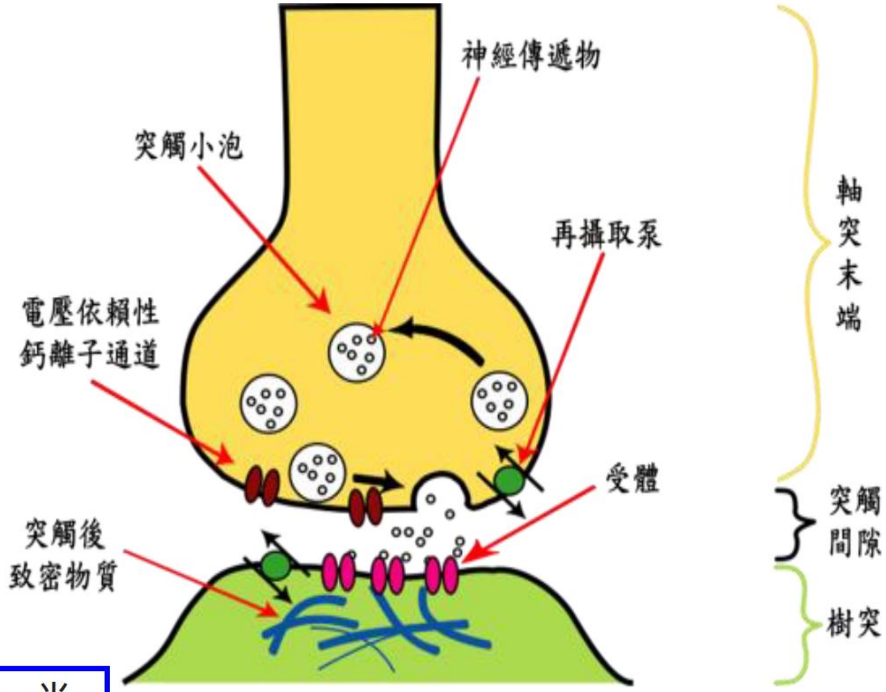
神經物質傳遞

神經物質傳遞循環

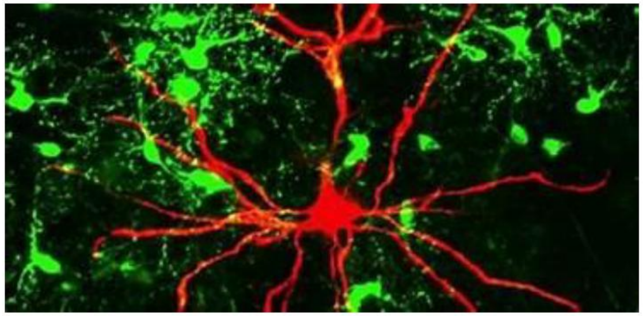
1.製造
2.傳輸
3.釋放
4.擴散
5.結合
6.觸動
7.分離
8.吸收
9.回復

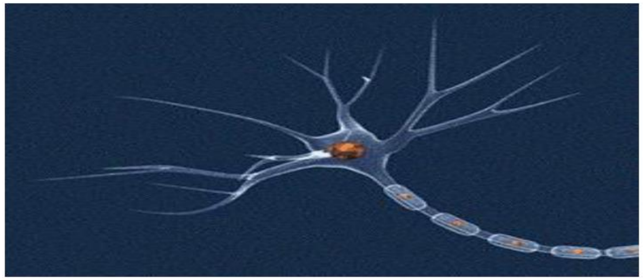
電線傳導速率為光速（3 ×10^8 m/s）一半
神經傳導速率 0.5 m/s ~200 m/s

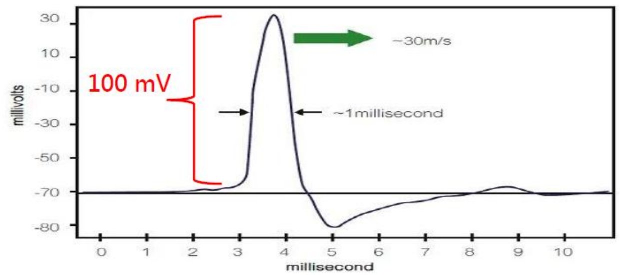




典型突觸的結構







腦中常見的神經傳遞物

神經傳遞物	腦中功用
乙醯膽鹼 (Acetylcholine，Ach)	興奮性神經傳遞物
γ-氨基丁酸 (GABA)	抑制性神經傳遞物
血清素 (Serotonin，5-HT)	睡眠-覺醒周期、心境食物和性行為、增強記憶力、防止老化
多巴胺 (Dopamine)	情緒、獎勵–激勵行為等
正腎上腺素 (Norepinephrine)	神經傳遞物、激素
腎上腺素(Epinephrine)	戰鬥或逃跑反應、瞳孔放大、血糖上升等
褪黑激素 (Melatonin)	調節生物鐘的激素
腦內啡 (Endorphin)	愉悅感

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取自台大心理系 黃從仁教授

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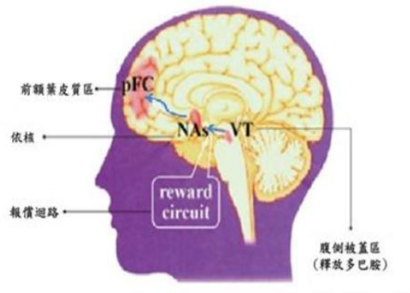
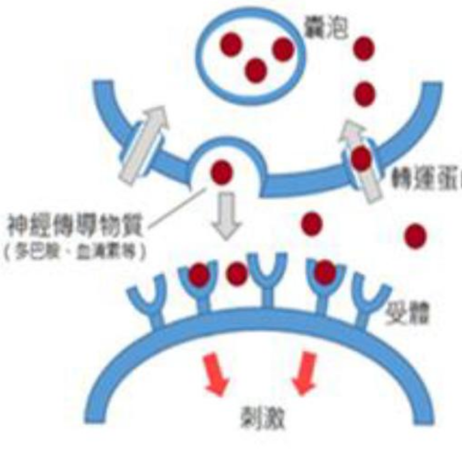
神經物質傳遞



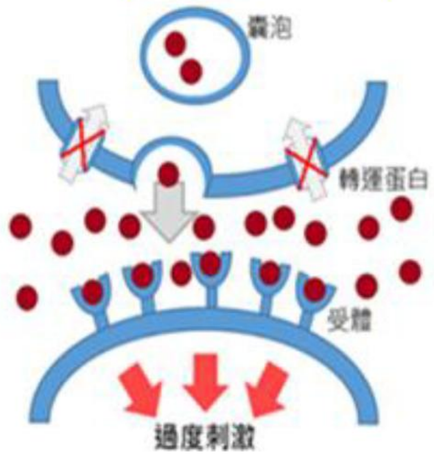
大腦與成癮相關圖

古柯鹼（Cocaine）是興奮劑，使用了會使人亢奮、加快心跳；
海洛因（Heroin）是鎮靜劑，可以做止痛之用。
二者對生理影響完全相反，
但它們造成的「成癮」機制卻相同，
都是作用在「報償系統」上。
殊途同歸——讓吸食者「成癮」。

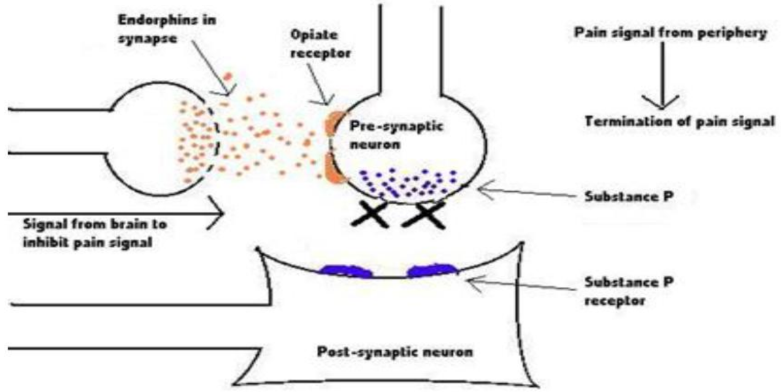
正常狀態



吸毒（Cocaine）狀態



嗎啡作用（Morphine）



酒精、嗎啡（Morphine）、鴉片（Opiate）

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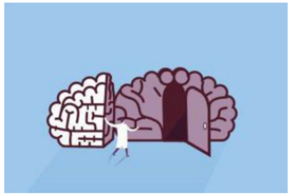
取自台大心理系 黃從仁教授

取自 國立清華大學生命科學系潘榮隆榮譽講座教授

結構、功能、連結

大腦的學習機制



資訊 (Information) → 感官 (Senses) → 大腦 (Brain)

大腦 (Brain)

突觸 (Synapses) → 短期記憶 (Short-term memory) → 工作記憶 (Working memory)

處理

比較

知道 / 經驗 (Know / Experience) → 長期記憶 (Long-term memory)

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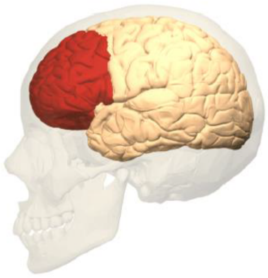
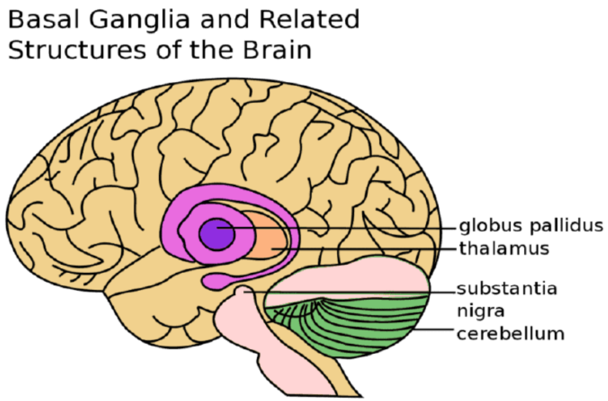
學習相關的大腦部位

Prefrontal cortex 前額葉皮層

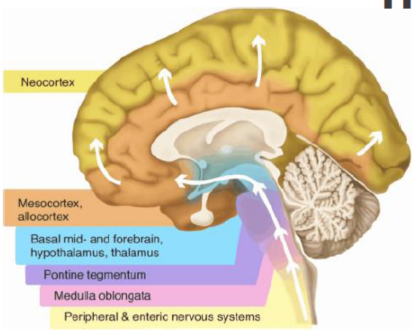
Short-term 短期記憶
working memory工作記憶



Basal ganglia 基底神經節



Neocortex 新皮層

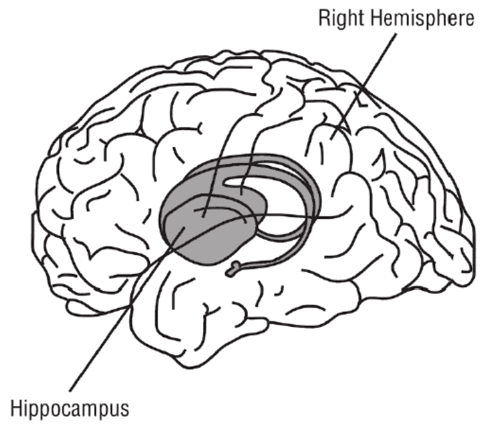


Information processor
訊息處理
Logics、Learning
Reasoning 邏輯、學習、
理性

Automatic or involuntary learning 自發或非自願學習
Implicit memories 內隱記憶

「學習」相關的大腦部位

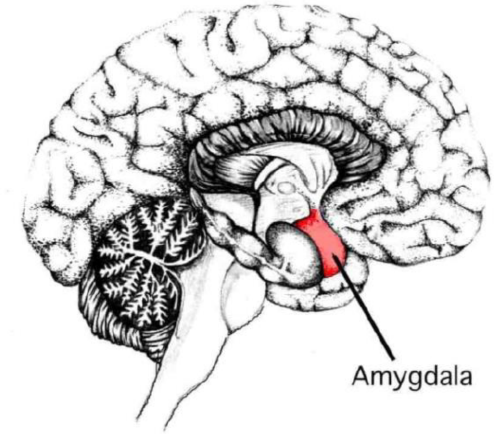
Hippocampus 海馬迴



Long-term episodic memories. 長期情景記憶

Short-term memory storage 短期記憶儲存

Amygdala 杏仁核



Creates sentimental and emotional responses；Helps create long-term episodic memories.

傷感情緒、協助長期情景記憶

記憶種類

感覺記憶 Sensory Memory

感覺訊息存在極短時間內（感覺登記），大約0.25秒到4秒之間。

短期記憶 Short-term Memory

少量資訊保持啟用狀態，在短時間內可使用。短期記憶的持續時間約5 - 20秒。資訊儲存容量約7 ± 2 個元素。

工作記憶 Working memory

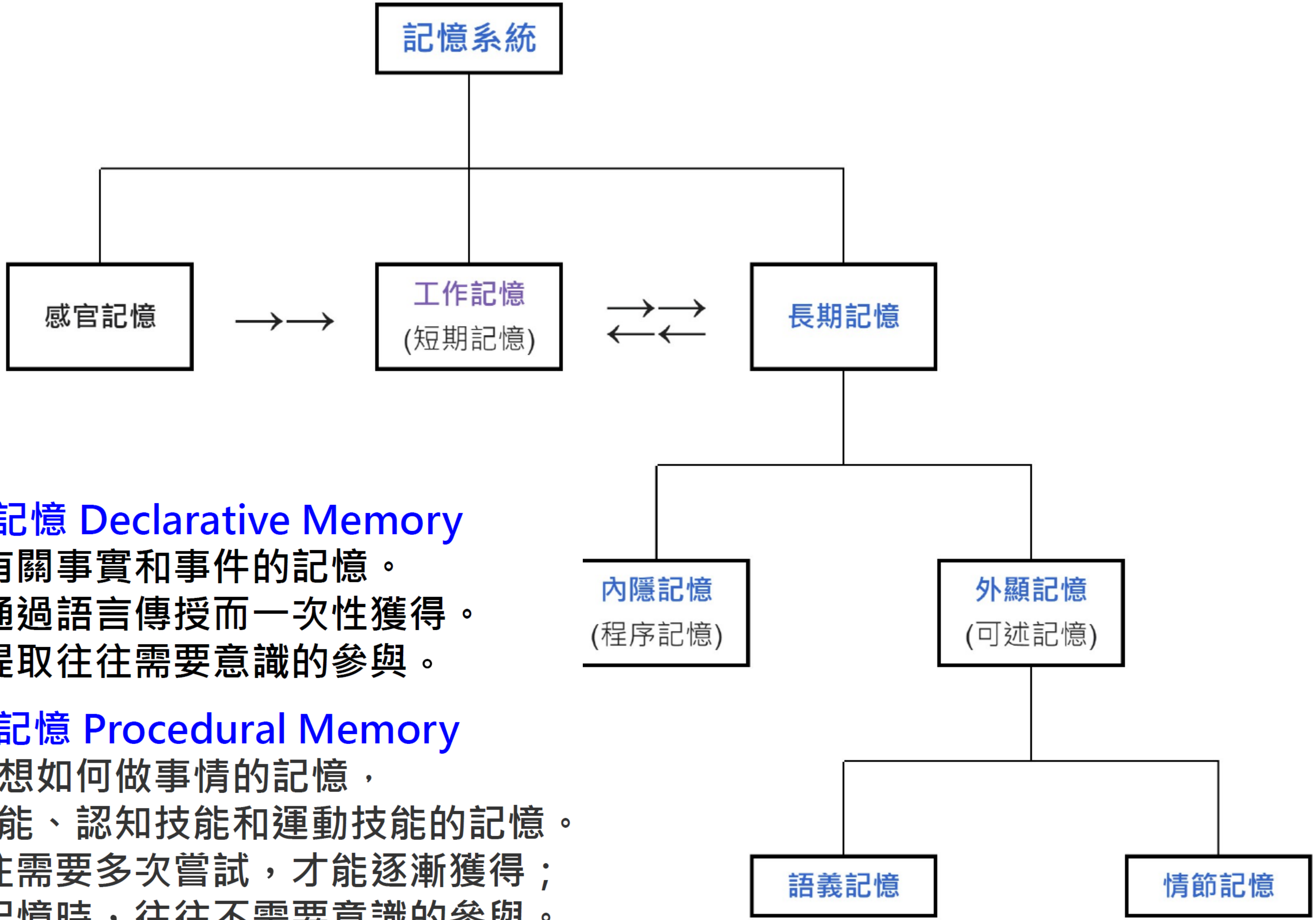
暫時保存資訊，以與長期記憶資料做比較，或推理、指導決策和行為。工作記憶允許對存儲資訊進行操作，短期記憶僅短期存儲資訊。

長期記憶 Long-term memory

能夠保持幾天到幾年、甚或終身的記憶。將短期記憶的神經連接強化、鞏固後，變為長期記憶。

（續）

記憶種類



陳述性記憶 Declarative Memory

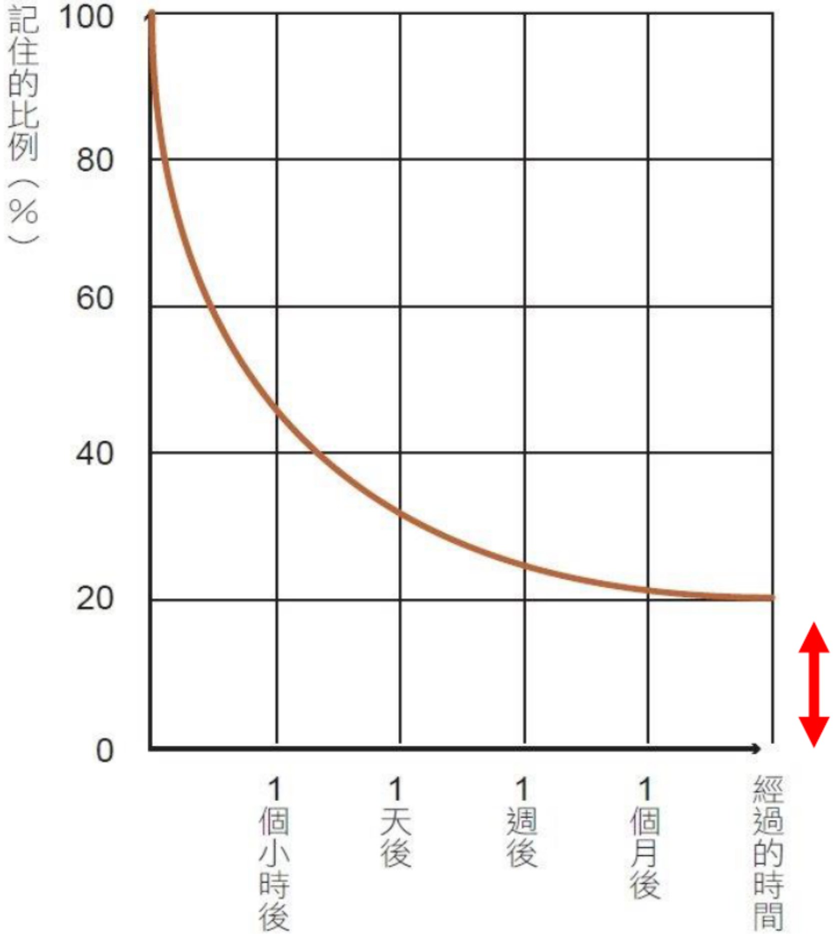
對有關事實和事件的記憶。
它可以通過語言傳授而一次性獲得。
它的提取往往需要意識的參與。

程式性記憶 Procedural Memory

回想如何做事情的記憶，
包括對知覺技能、認知技能和運動技能的記憶。
這類記憶往往需要多次嘗試，才能逐漸獲得；
在利用這類記憶時，往往不需要意識的參與。

遺忘曲線

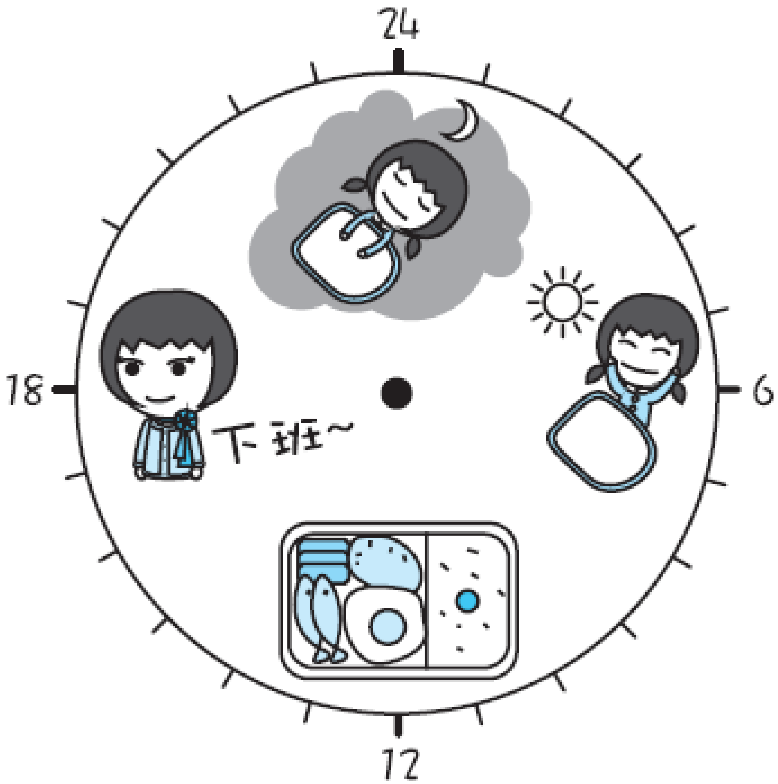
赫爾曼·艾賓浩斯
（Hermann Ebbinghaus）
1850~1909
德國心理學家



Ebbinghaus Forgetting Curve.
（Source：Schaefer, 2015）

腦力有限，應保養頭腦

人體的生理活動、體溫變化、睡眠、荷爾蒙分泌等，會依著晝夜的改變，循環變化，稱為**「晝夜節律」(circadian rhythms)**。

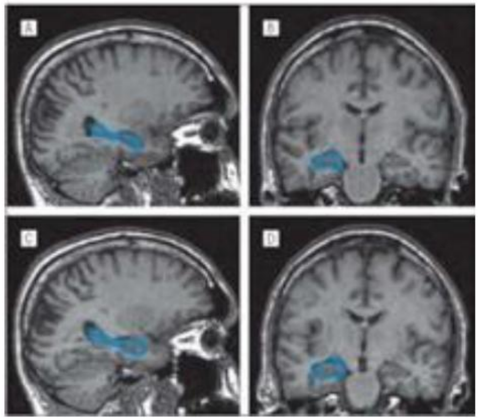


Six ways to grow your hippocampus

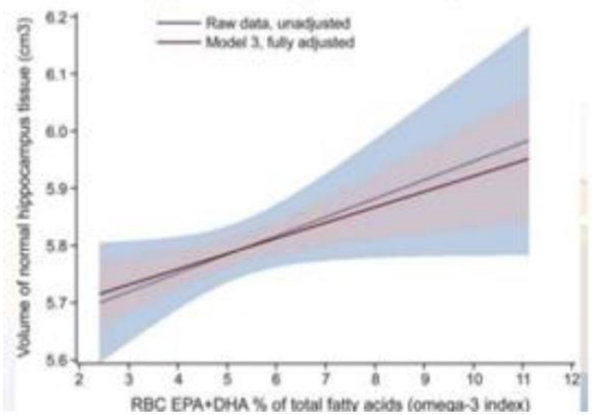
1. 運動

前

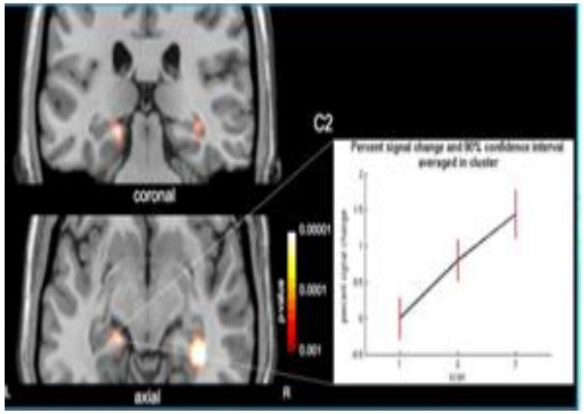
後



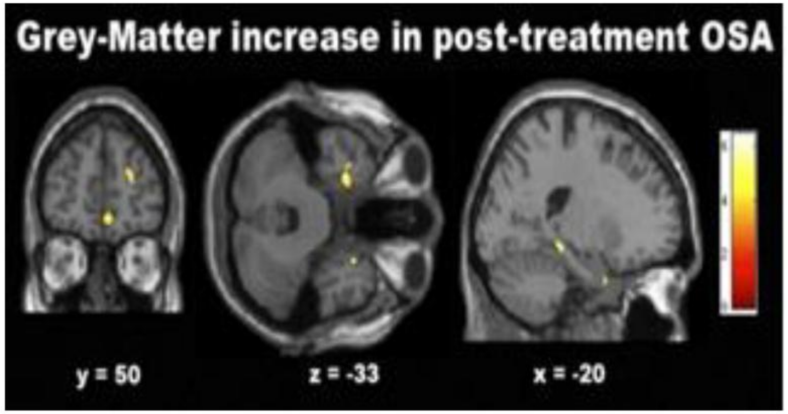
2. Ω-3 脂肪酸



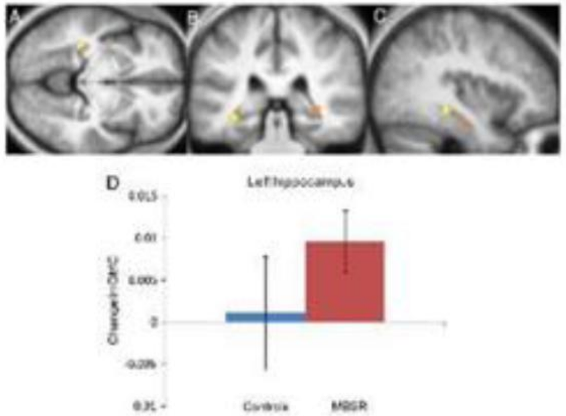
3. 學習



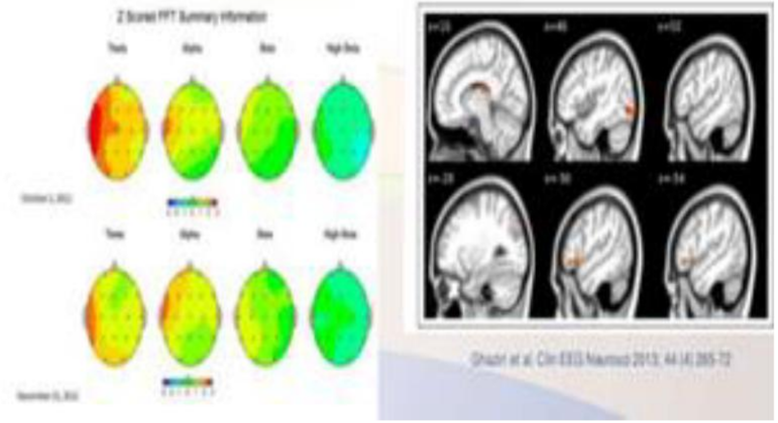
4. 充足睡眠



5. 好的信仰與靈修



6. 神經回饋（刺激）



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取自台大心理系 黃從仁教授

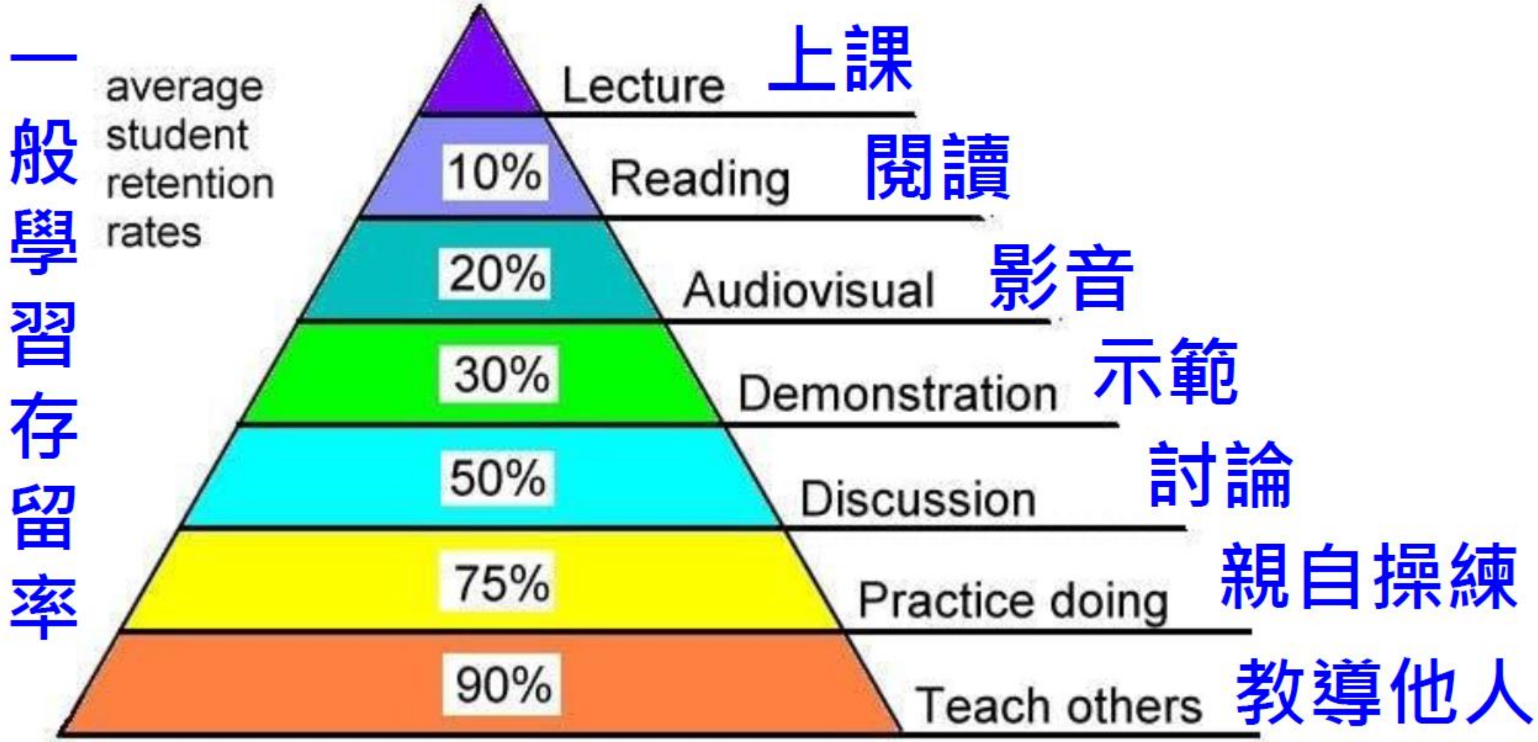
取自 國立清華大學生命科學系潘榮隆榮譽講座教授

注意事項

6. 最有效的學習法：教導其他人

學習金字塔

Learning Pyramid

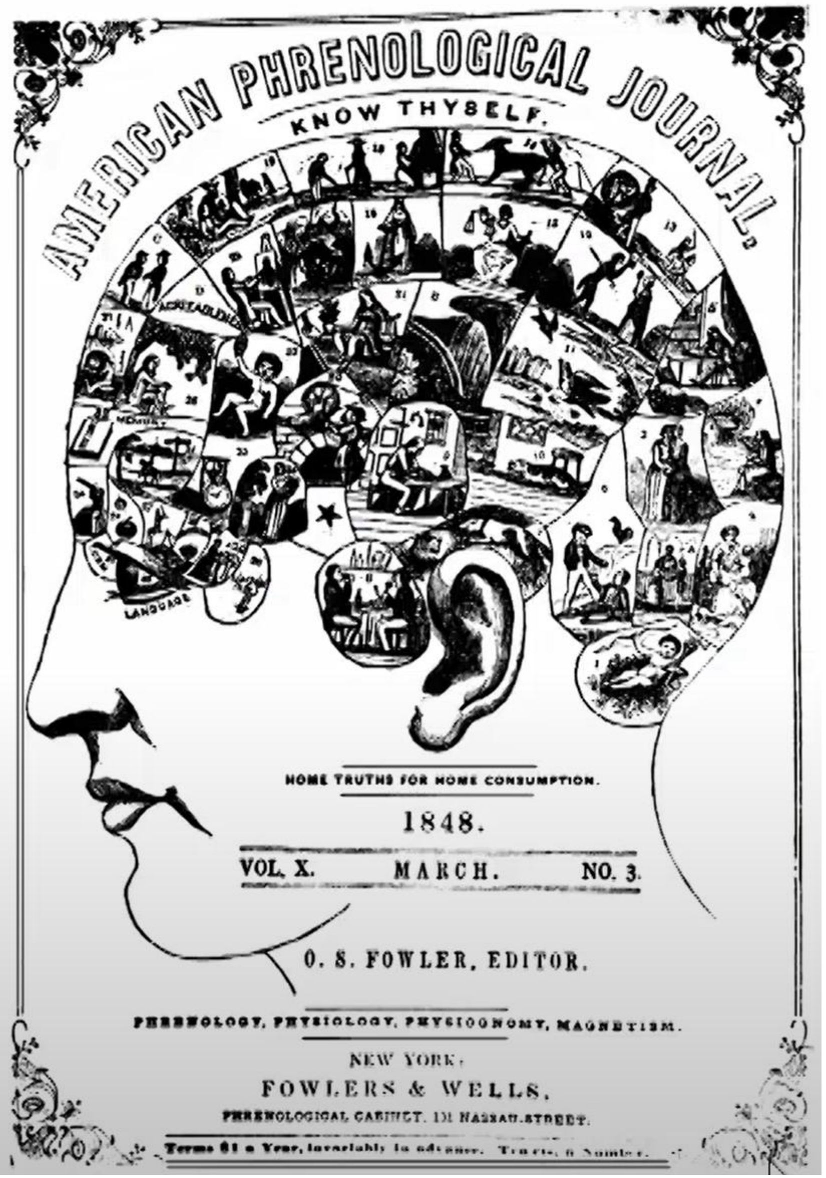


Source: National Training Laboratories, Bethel, Maine

Phrenology (顱相學)

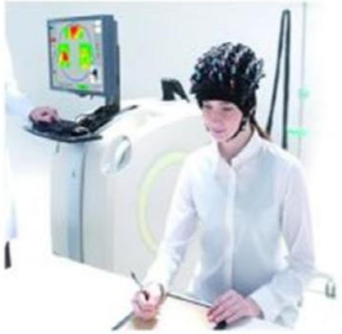
- ✓ Franz Joseph Gall (1810):
  The brain is composed of a set of mental organs each involved in a different psychological function or trait.

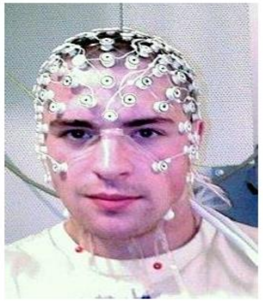
- ✓ “Location of the Mental Organs



認識大腦「結構」、「功能」、「改變」…

認識研究大腦的「儀器」





功能性近紅外光譜
(Functional near infrared spectroscopy / fNIRS)

功能性核磁共振儀
(Functional Magnetic Resonance Imaging / fMRI)

腦電造影、腦波量測
(Electroencephalography / EEG)

單光子輻射電腦斷層掃描
(Single Photon Emission
Computed Tomography / SPECT)

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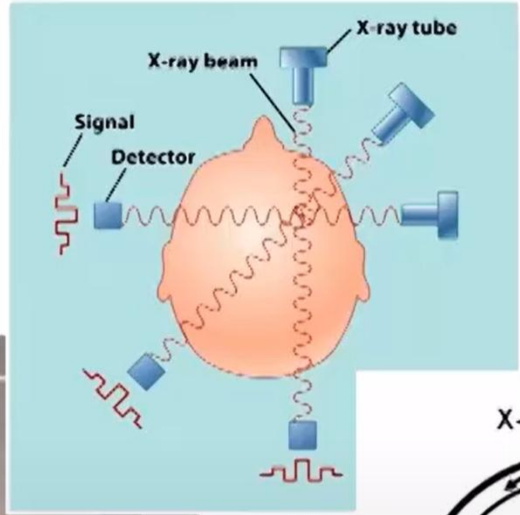
Computed tomography (CT)

- Projection of the X-ray from multiple angles
- 3D images based on tissue density



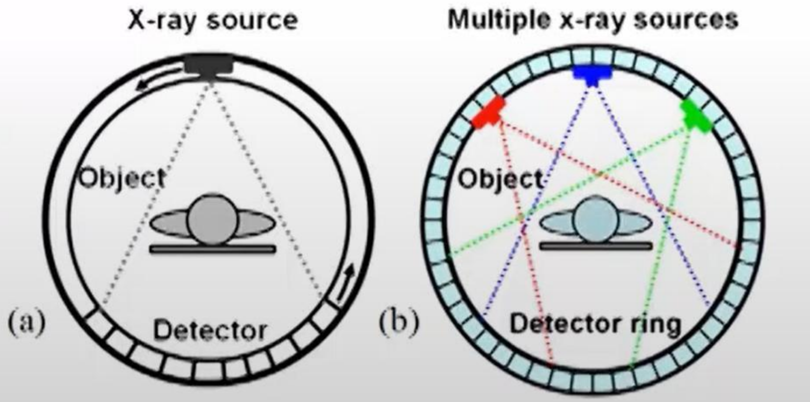


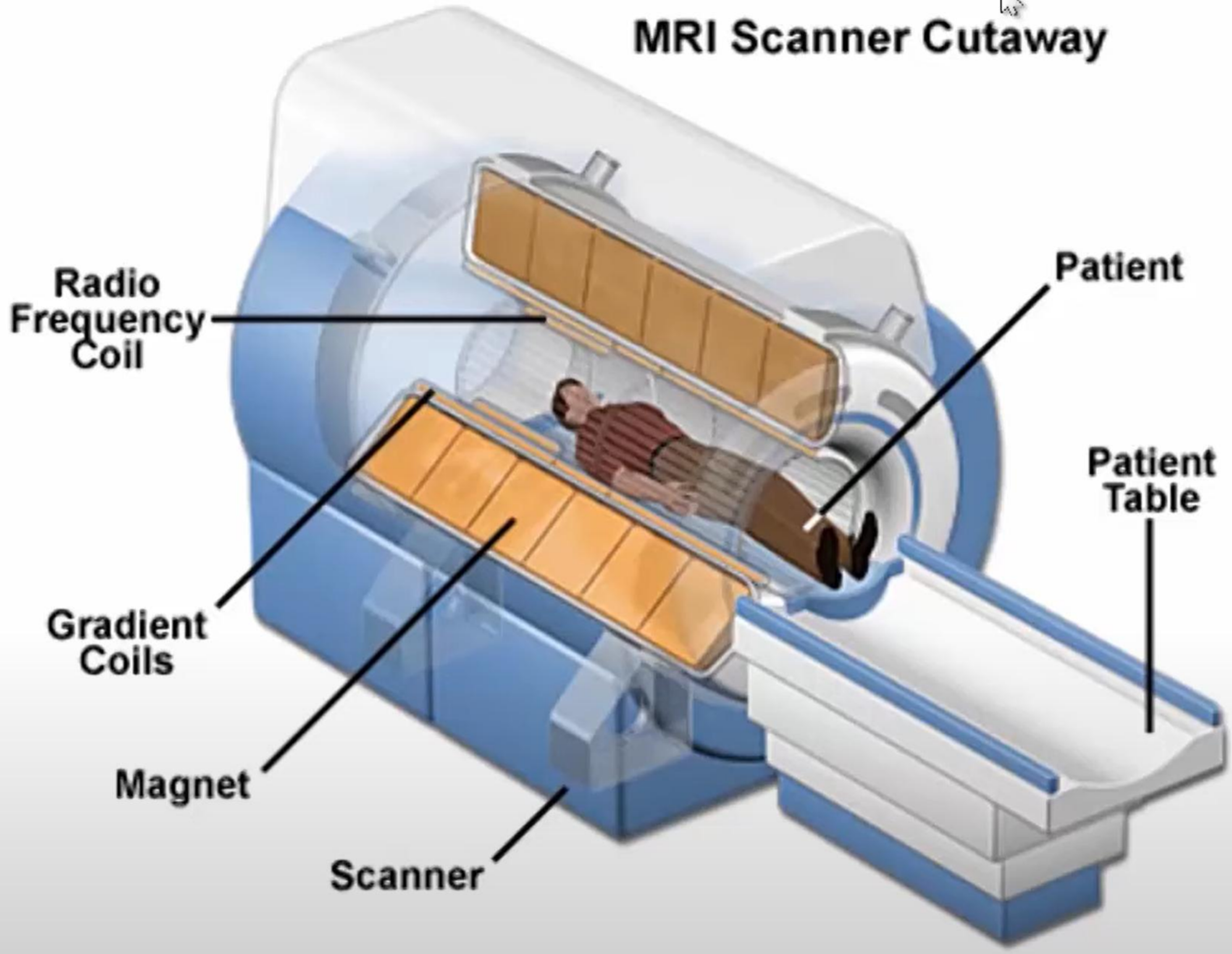
Figure 1: (a) Configuration of the conventional third generation CT scanner. (b) The multiplexing CT scanner enables simultaneous collection of multiple projection images due to the multiplexing principle.

Magnetic Resonance Imaging (MRI)

- MRI = NMR + Imaging
- Magnetic: Magnetization induced by magnetic field
  - 訊號的來源，人體中小磁鐵的磁化
- Resonance: Excite and receive of magnetization
  - 小磁鐵訊號的偵測
- Imaging: Spatial encoding of magnetization
  - 訊號轉為影像的方式(空間編碼)

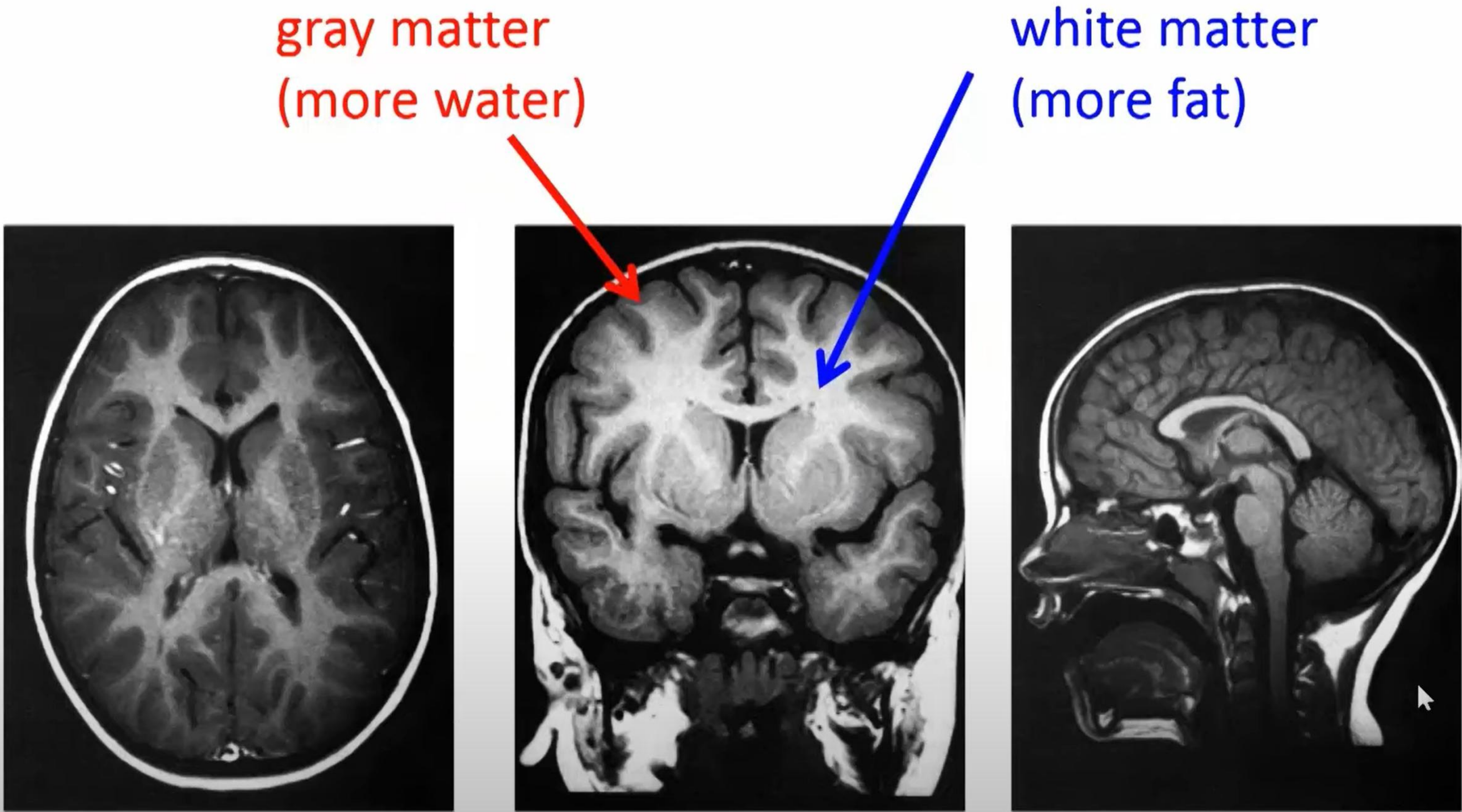
Necessary equipment: 3 Tesla MRI scanner

60人教室，約20坪



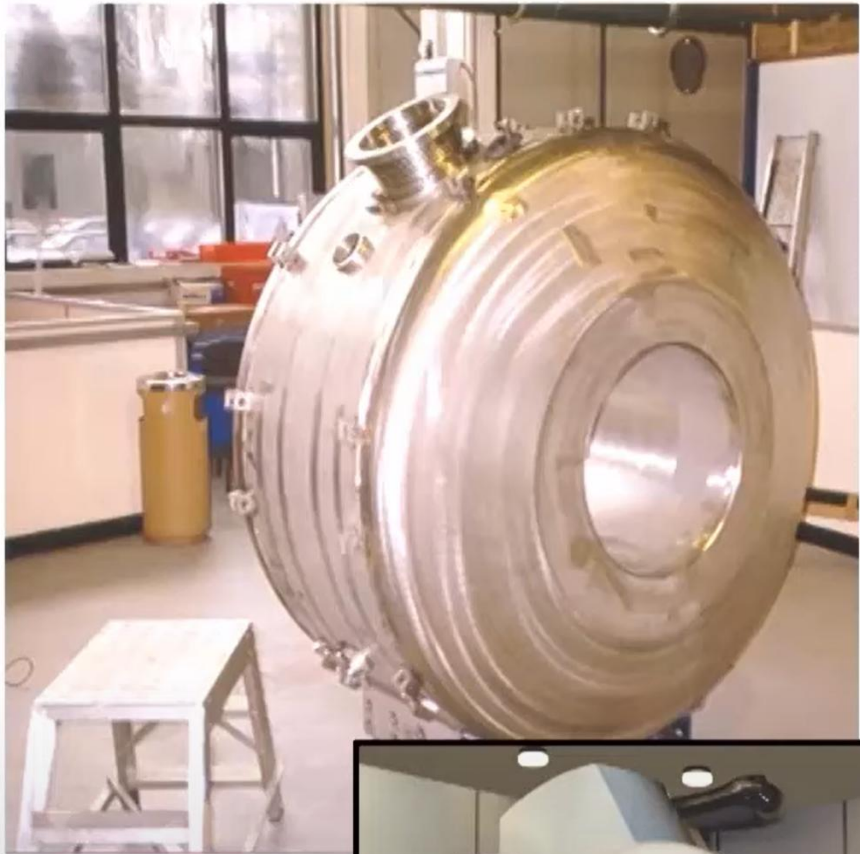
http://www.magnet.fsu.edu/education/tutorials/magnetacademy/mri/fullarticle.html

High-resolution T1 structural MRI images



Magnet

- Earth's magnetic field = 0.5 Gauss
- 1 Tesla = 10,000 Gauss
- 3 Tesla (T) =
  3 X 10,000 /0.5 = 60,000
  Earth's magnetic field

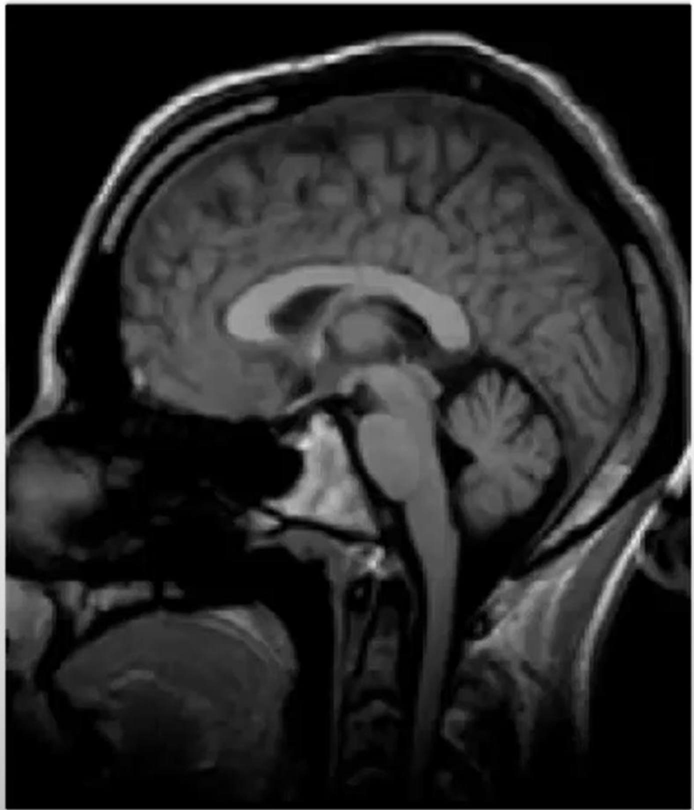


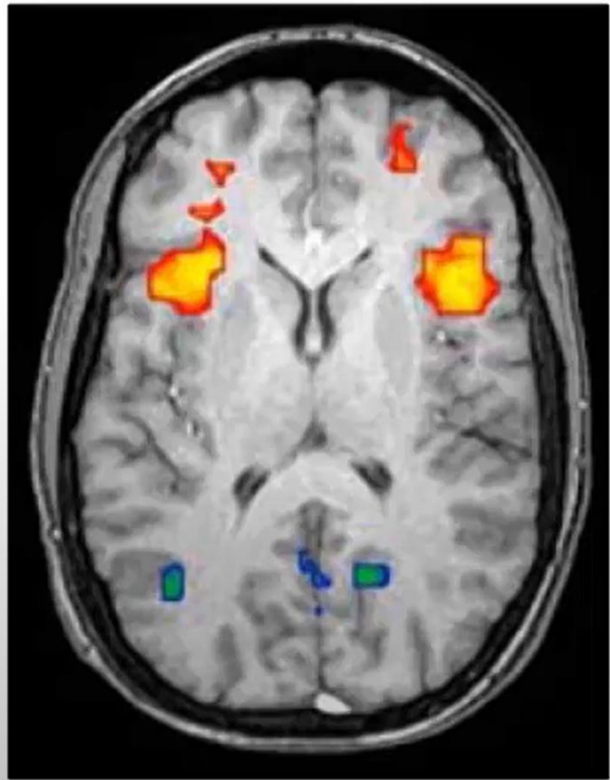
- Permanent
- Superconducting
- Active shielding

MRI safety

MRI vs. fMRI

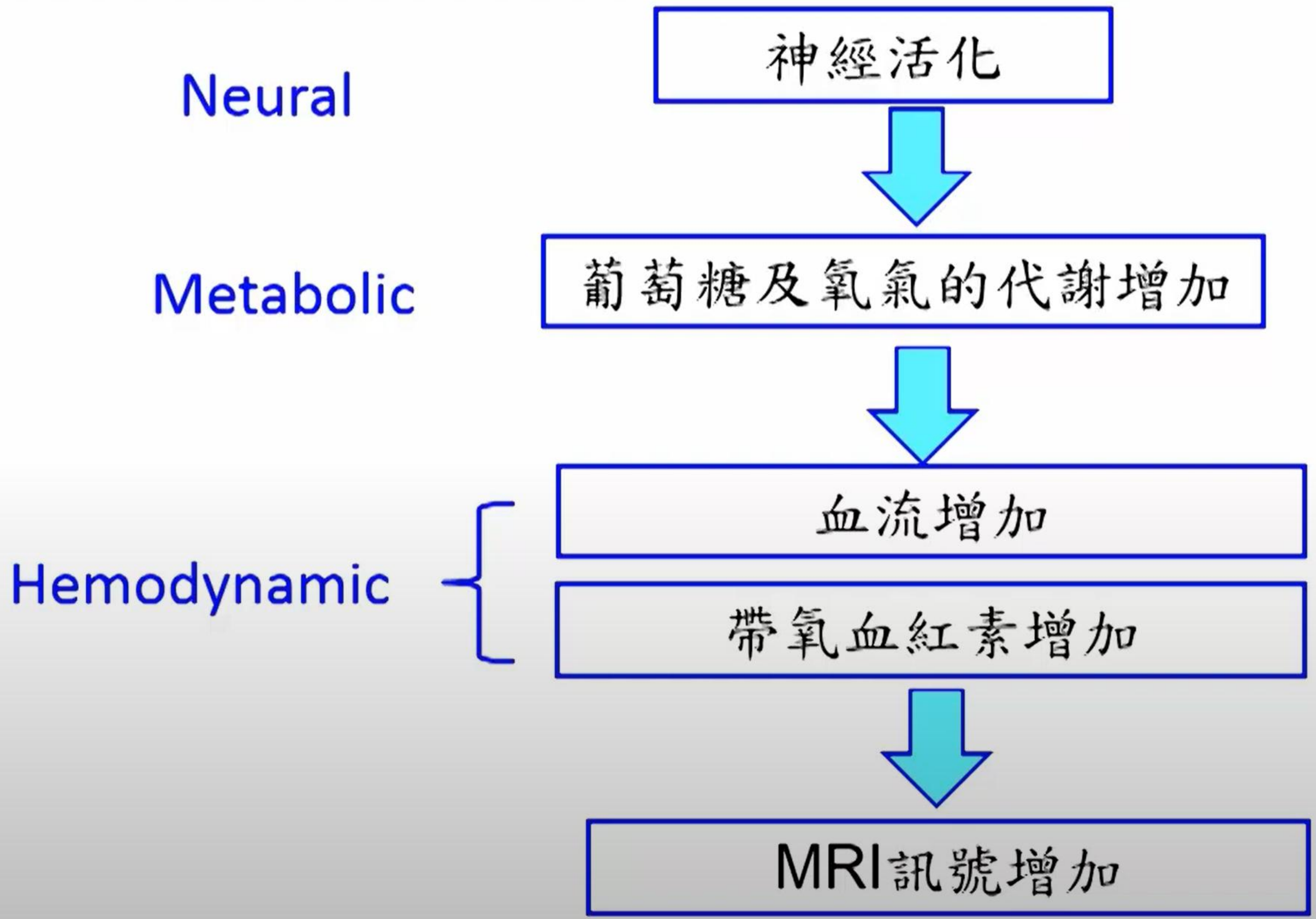
- MRI studies brain anatomy
- Functional MRI (fMRI) studies brain function





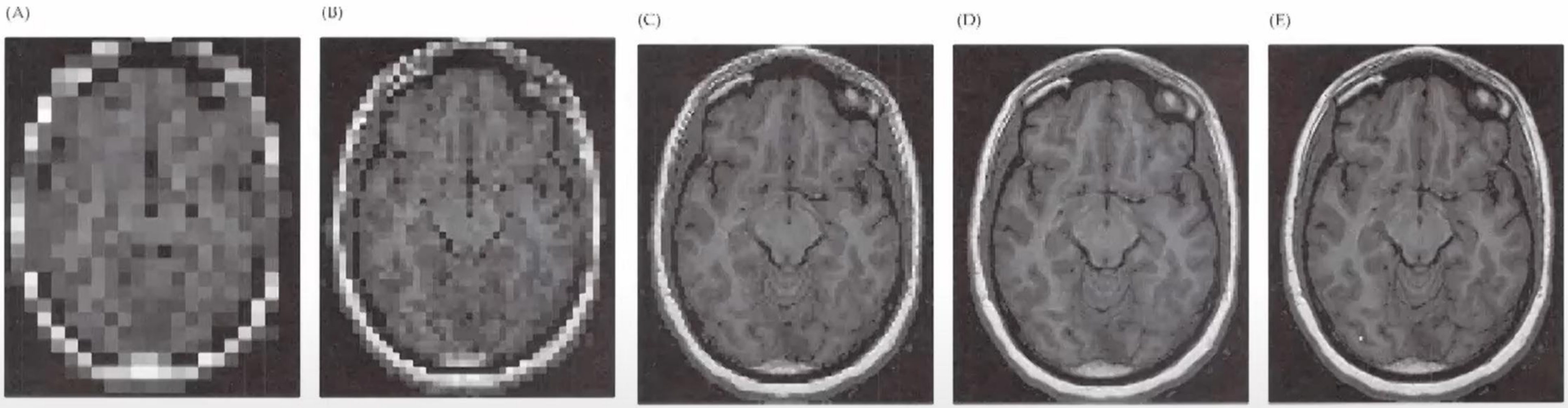
fMRI: BOLD signal

- Blood Oxygenation Level Dependent (BOLD) signal



From mathematics to physics

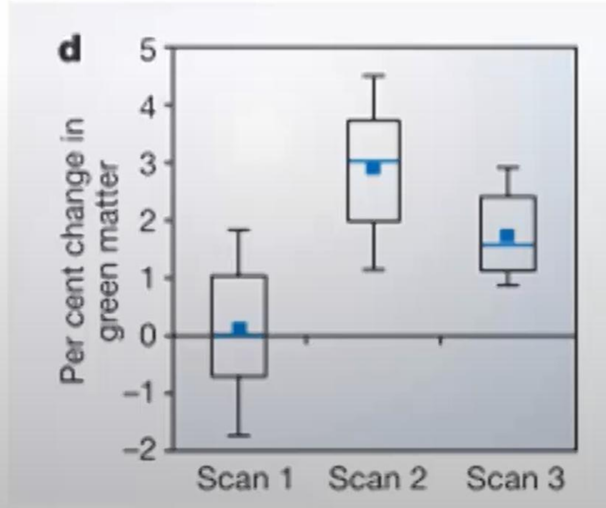
- A digital image = a mathematical matrix
- A pixel in the matrix has a "physical" dimension (e.g. mm)



8mm 4mm 2mm 1.5mm 1mm

Structural MRI application: Learning and memory

- Changes in gray matter induced by training:
  - Students inexperience at juggling
  - Divided into two groups (scan them)
  - One groups practices juggling for 3 months, and the other doesn't (scan again)
  - Wait 3 months, no one juggles (scan again)



MRI scanner and fMRI scan

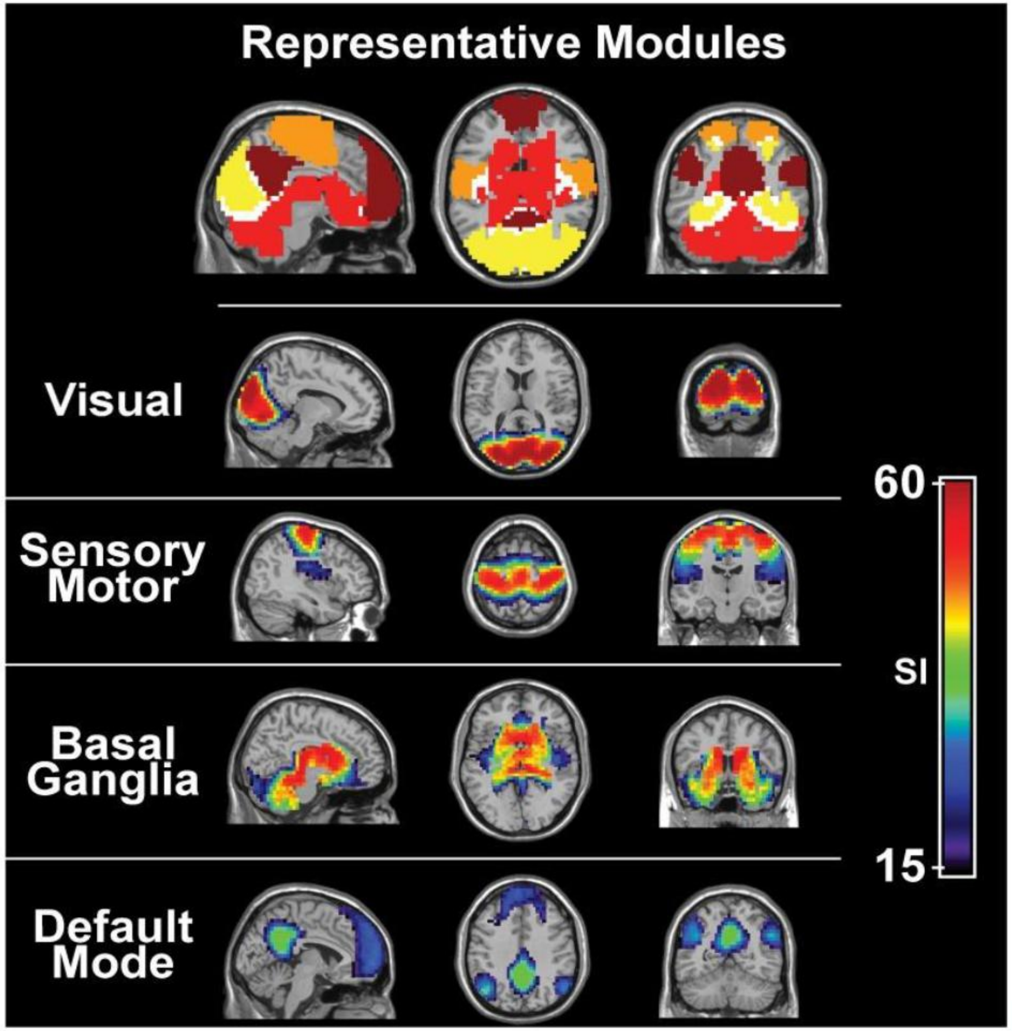
在靜息態下，
視覺系統及體感系統仍然活躍

視覺

體感運動

基底神經節

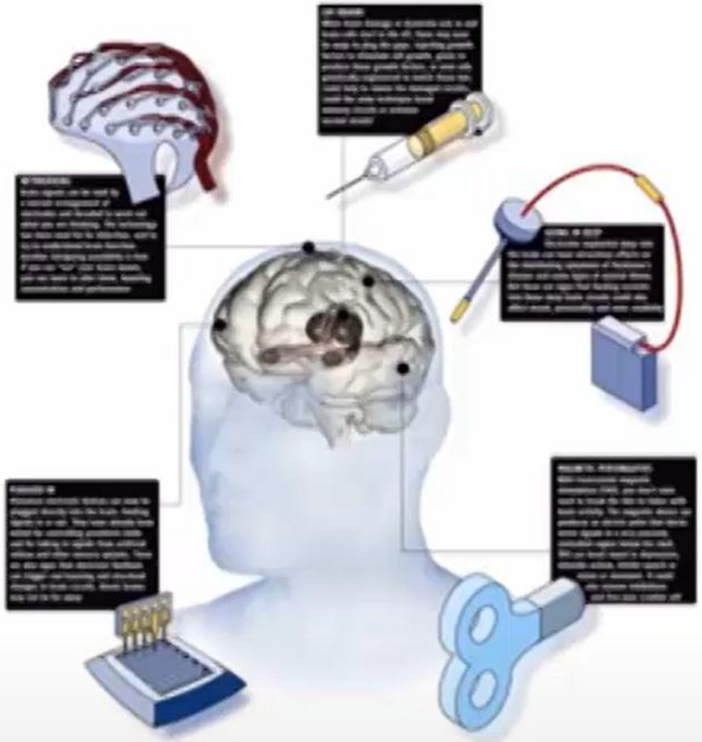
預設模式



（維基百科 https://reurl.cc/QeZ5e0 ）

Methods of cognitive neuroscience

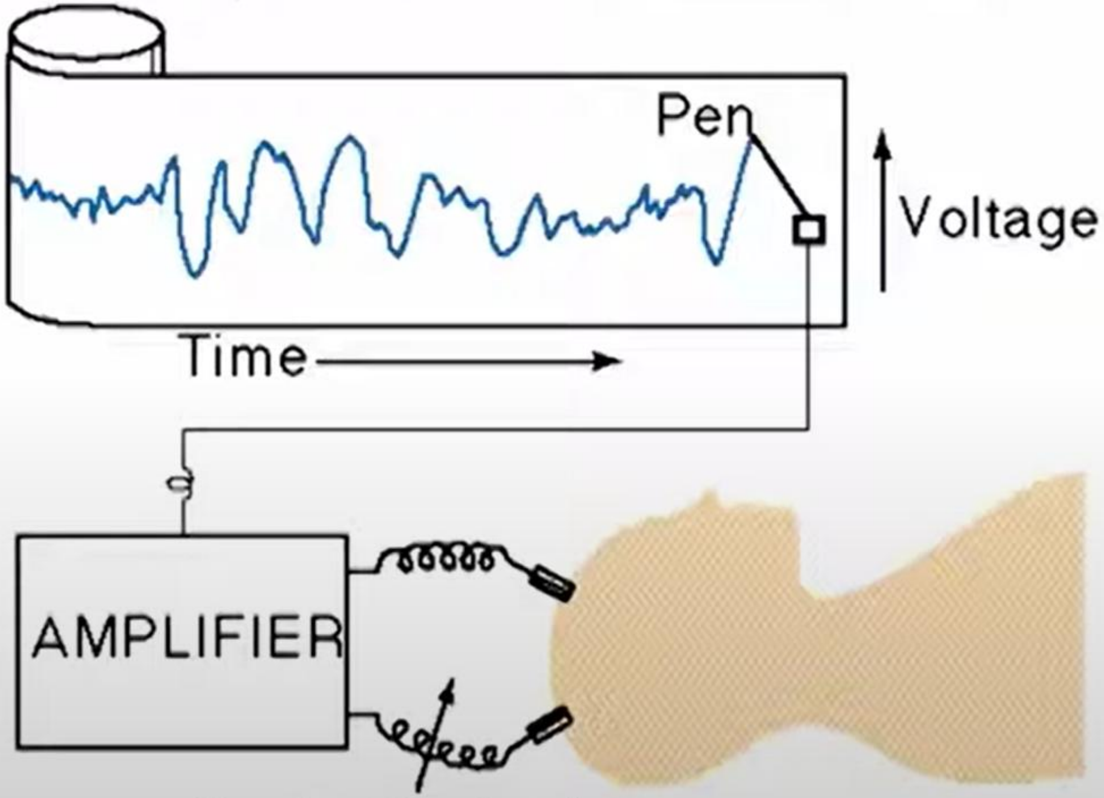
- Electrophysiology:
  - Electroencephalography (EEG)
  - Magnetoencephalography (MEG)
  - Transcranial magnetic stimulation (TMS)
- Neuroimaging:
  - Magnetic Resonance Imaging (MRI)
  - functional MRI (fMRI)
  - Positron emission tomography (PET)



Electroencephalography (EEG)

- **ElectroEncephaloGraphy:** a recording (graphy) of electrical signal (electro) from the brain (Encephalo).

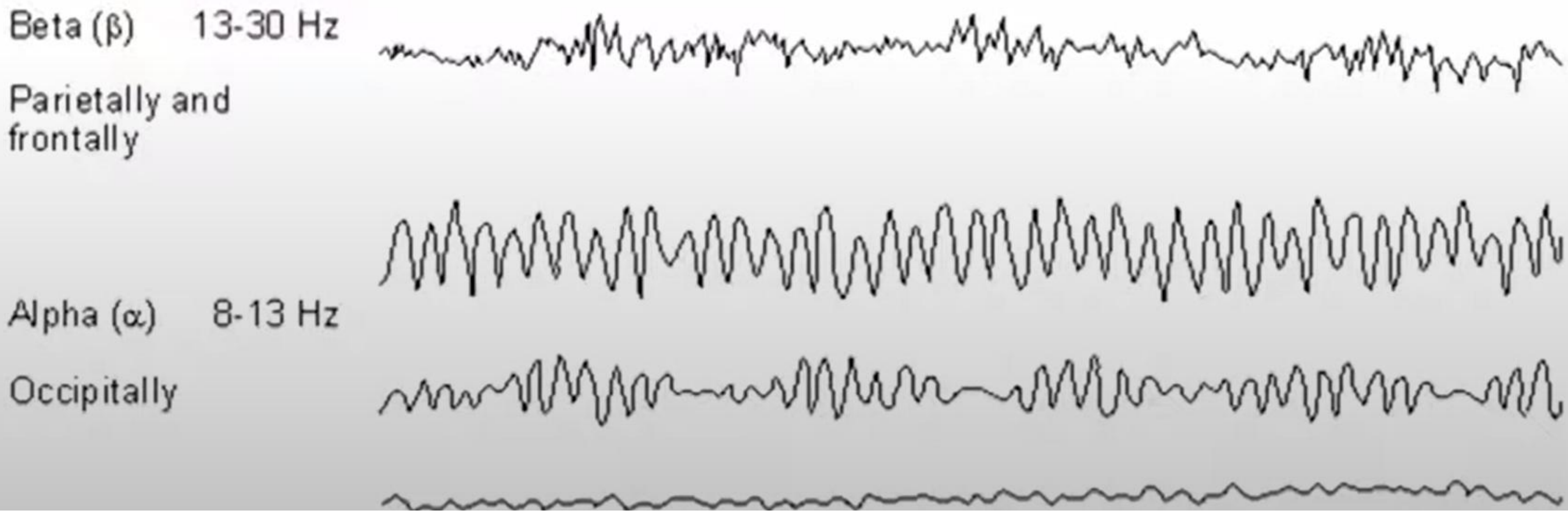
Electroencephalography Recording System



Electroencephalography (EEG)

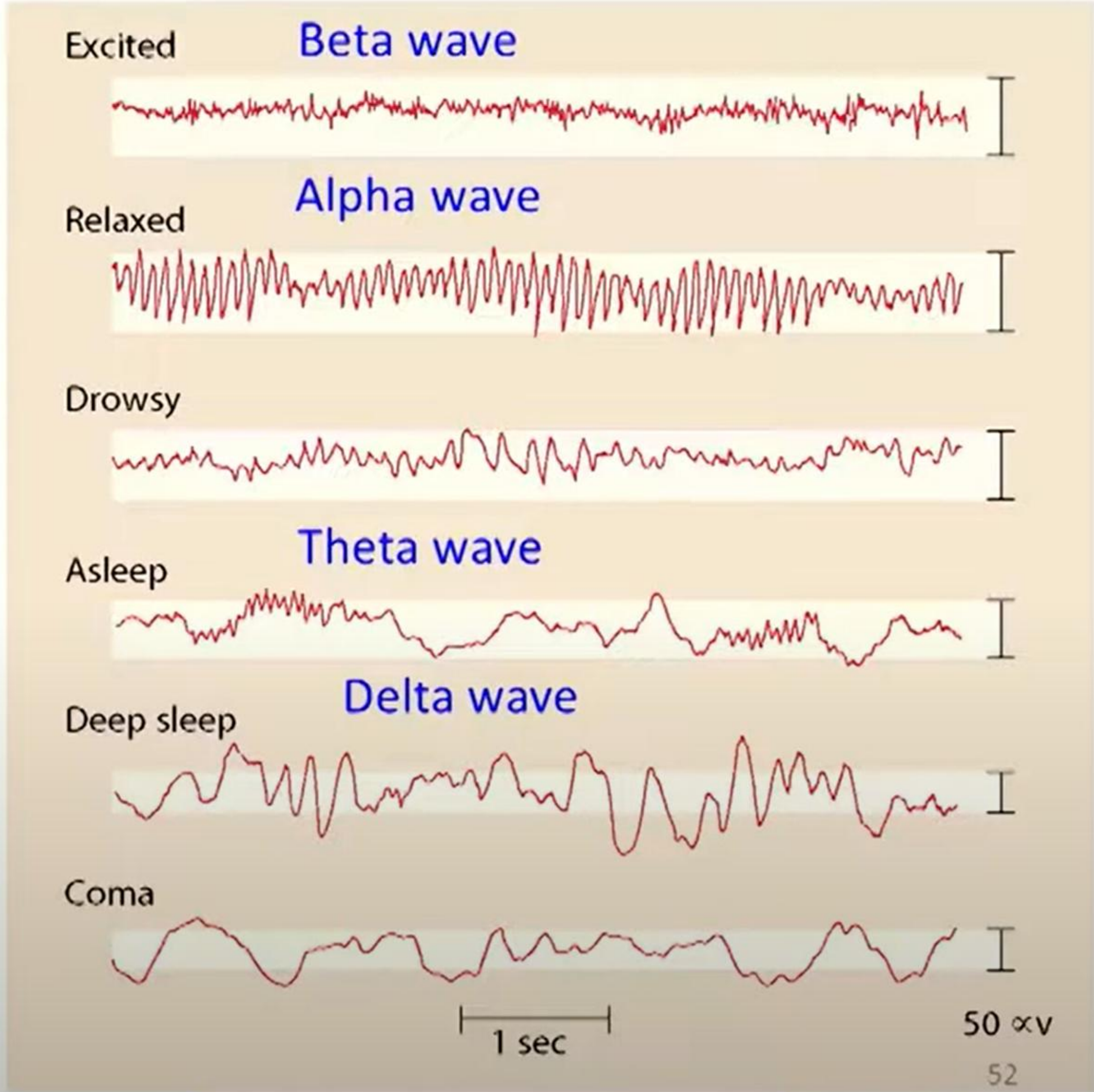
- Electrical activity is generated by the activity of billions of neurons.
- This electrical activity combines together and produces electrical fields which propagate through the brain and skull and can be measured with scalp electrodes.

- Berger’s observations revealed the basic phenomenology of human EEG:
  - ➢ Recorded brain activity changes according to the functional state of the brain: Sleep, hypnosis, and pathological states (epilepsy)
  - ➢ Frequency and amplitude

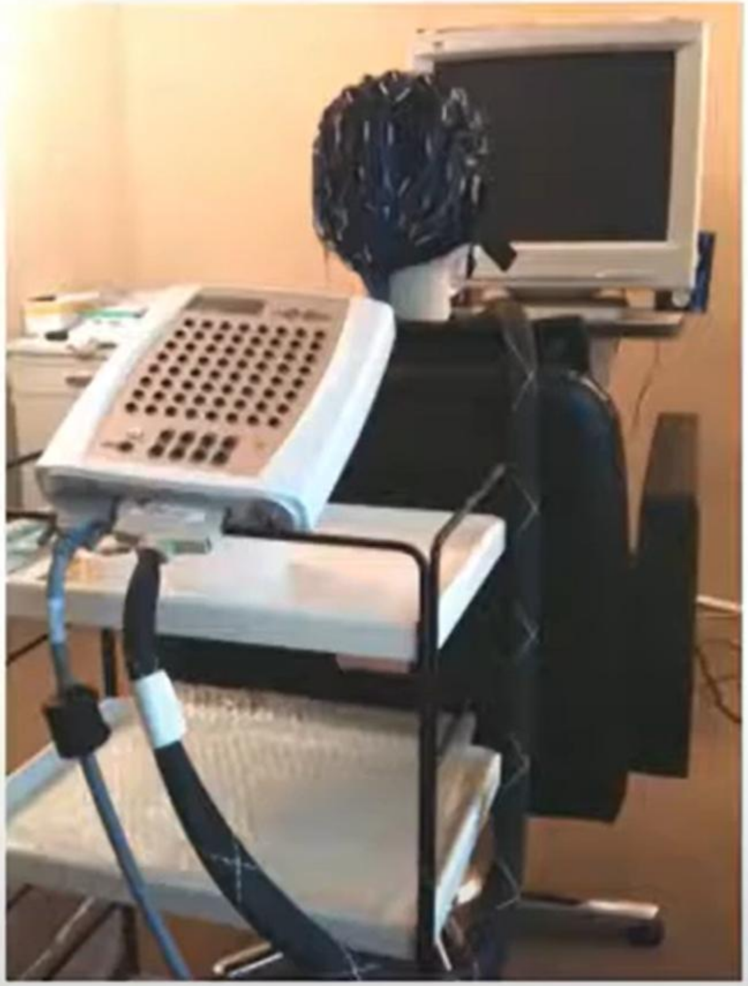


EEG in the states of vigilance

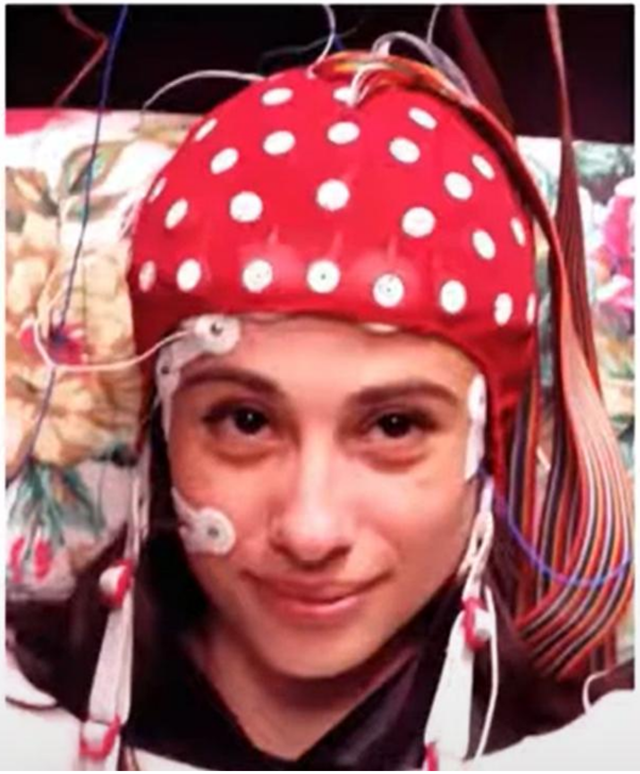
- Frequency Ranges:
  - ➢ Beta wave : 13-30 Hz
  - ➢ Alpha wave: 8-13 Hz
  - ➢ Theta wave: 4-8 Hz
  - ➢ Delta wave : 1-4 Hz



EEG recording system

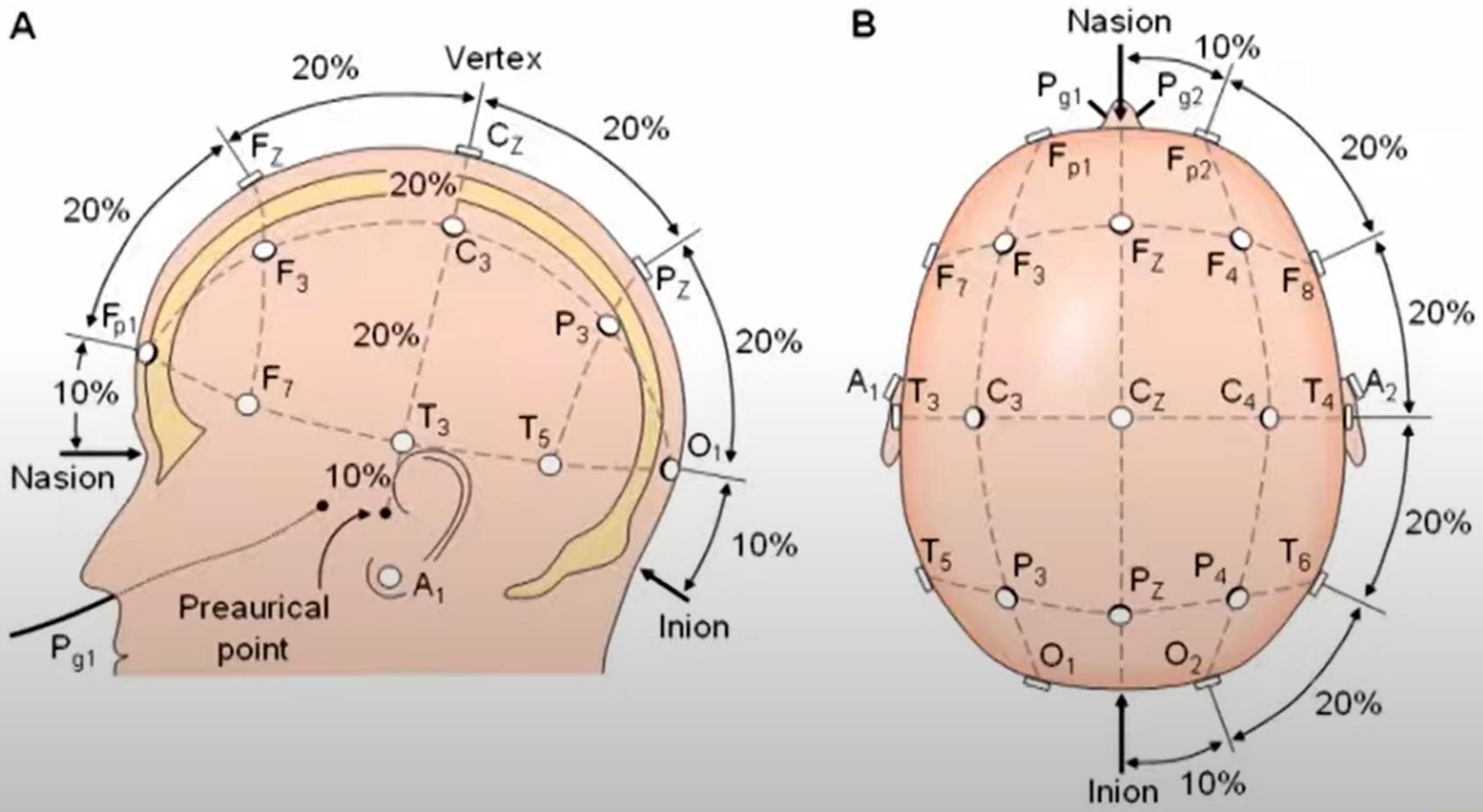


Electrode cap



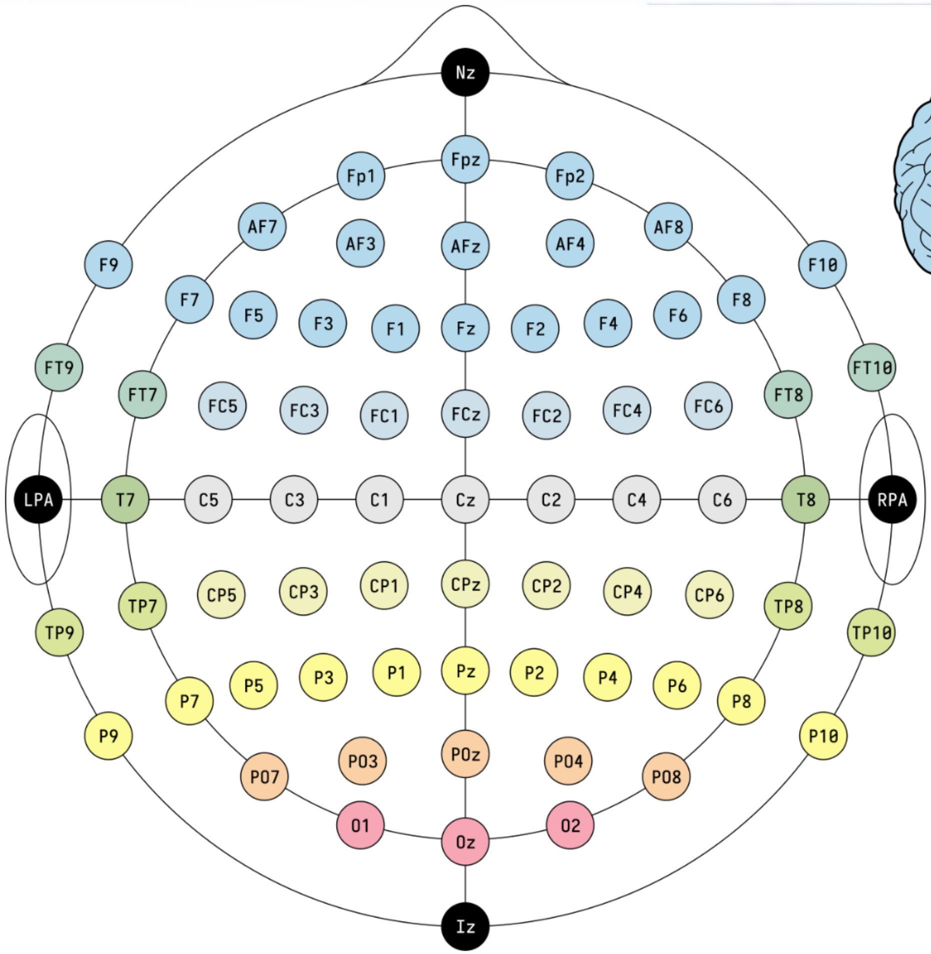
10-20 international system of electrode placement

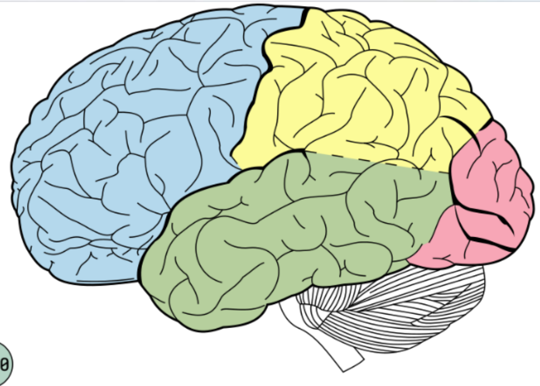
- Nasion: point between the forehead and the skull
- Inion: bump at the back of the skull

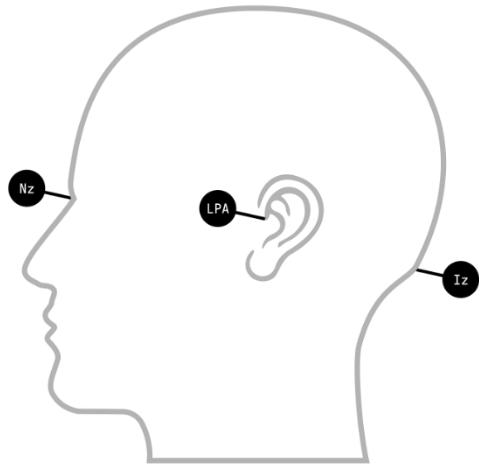


10-20 international system of electrode placement

- Montage
- Location: Frontal, Temporal, Parietal, Occipital, Central, Z for the central line







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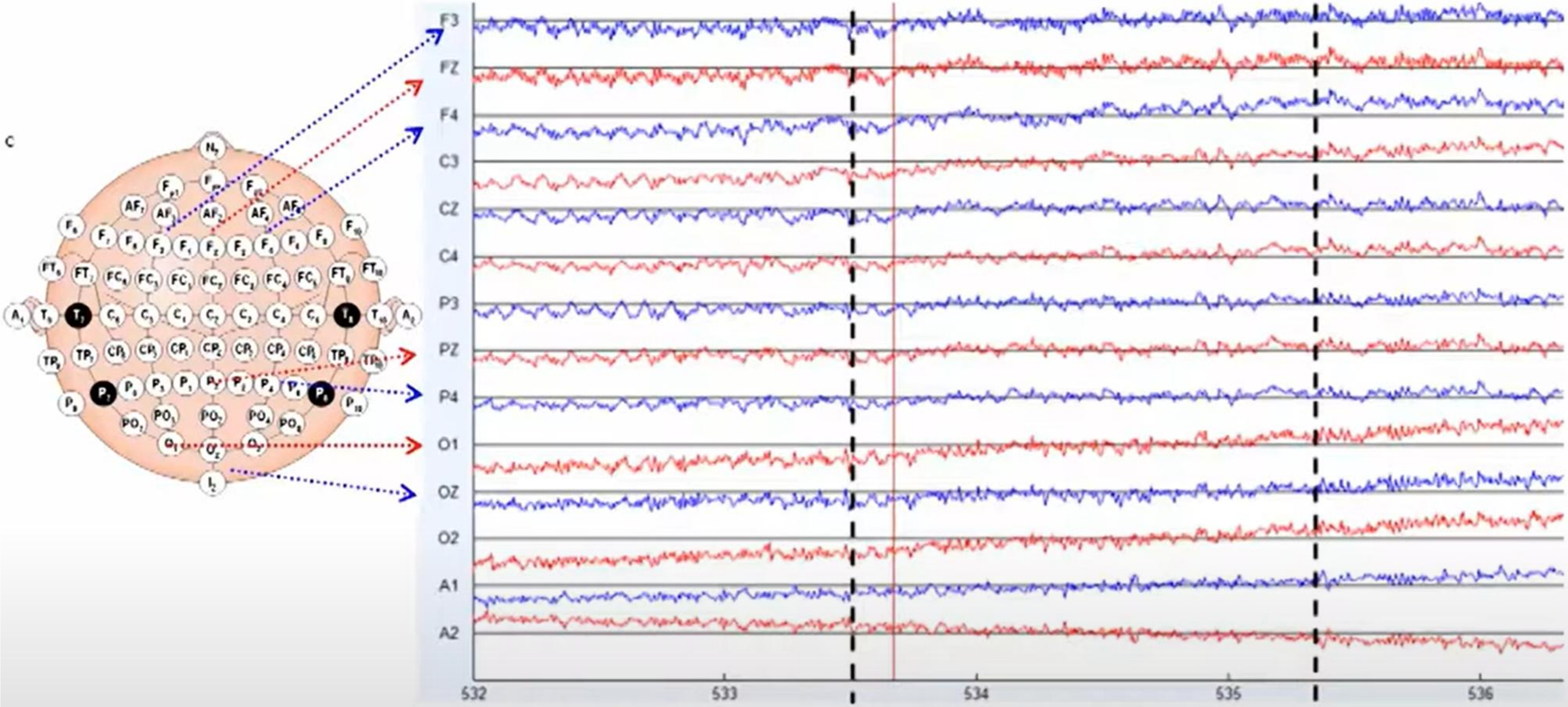
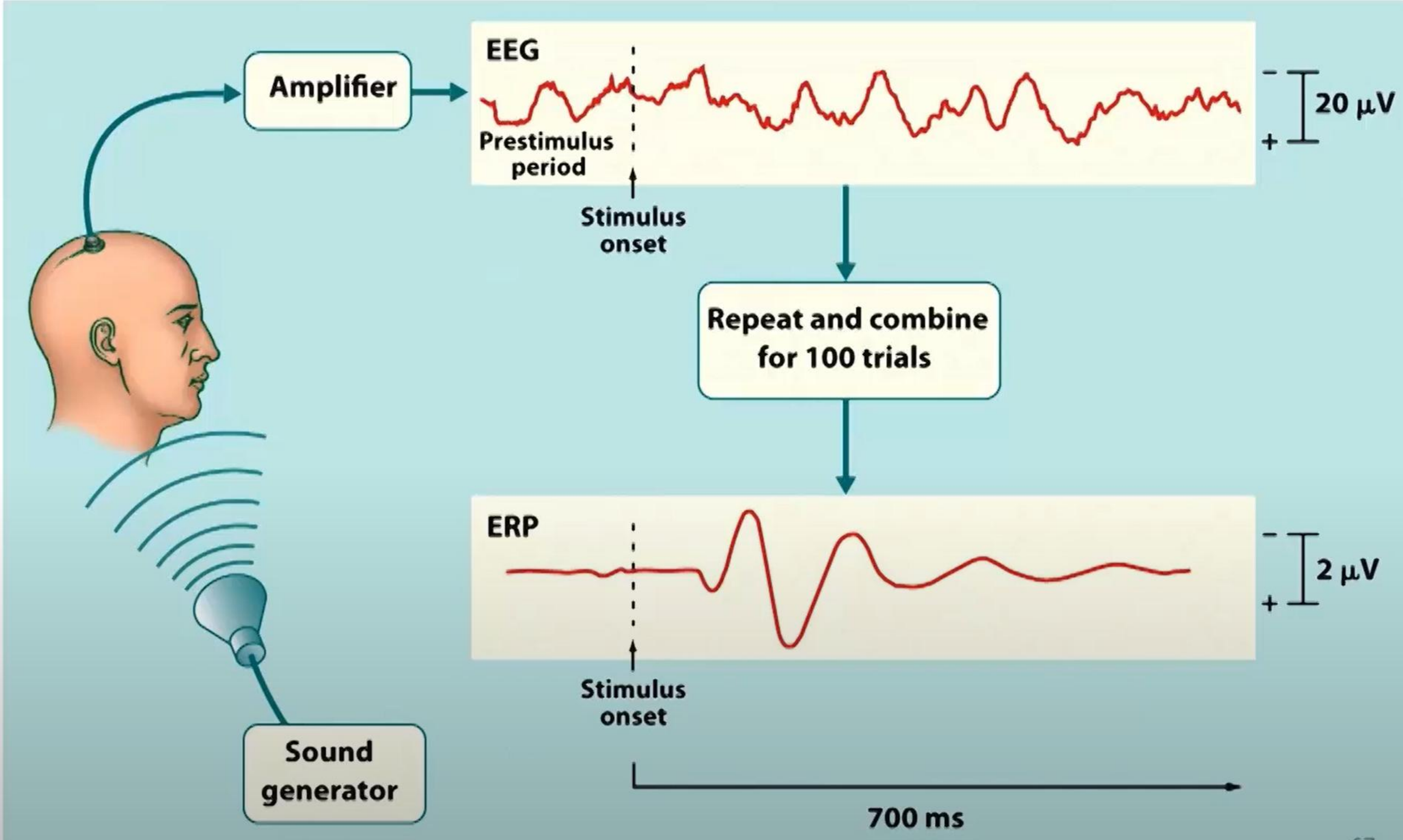
EEG channels



- channel: recording from a pair of electrodes
- Multichannel EEG recording: up to 40 channels recorded in parallel

From EEG to event-related potentials (ERPs)



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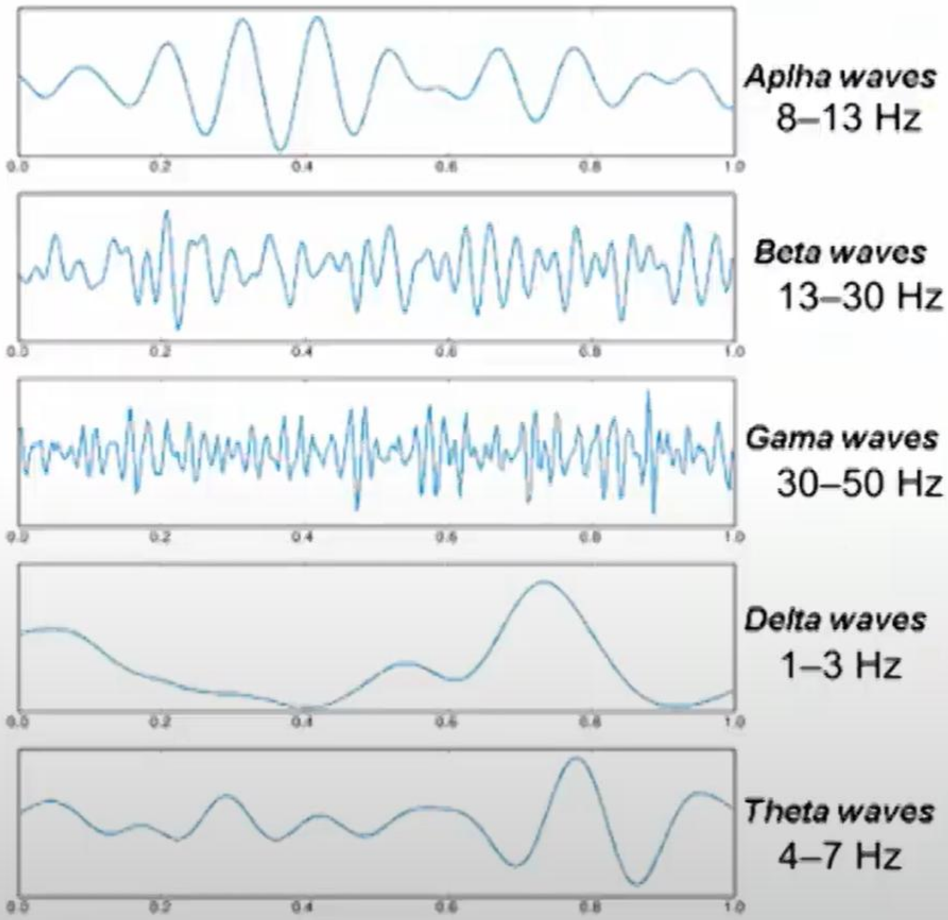
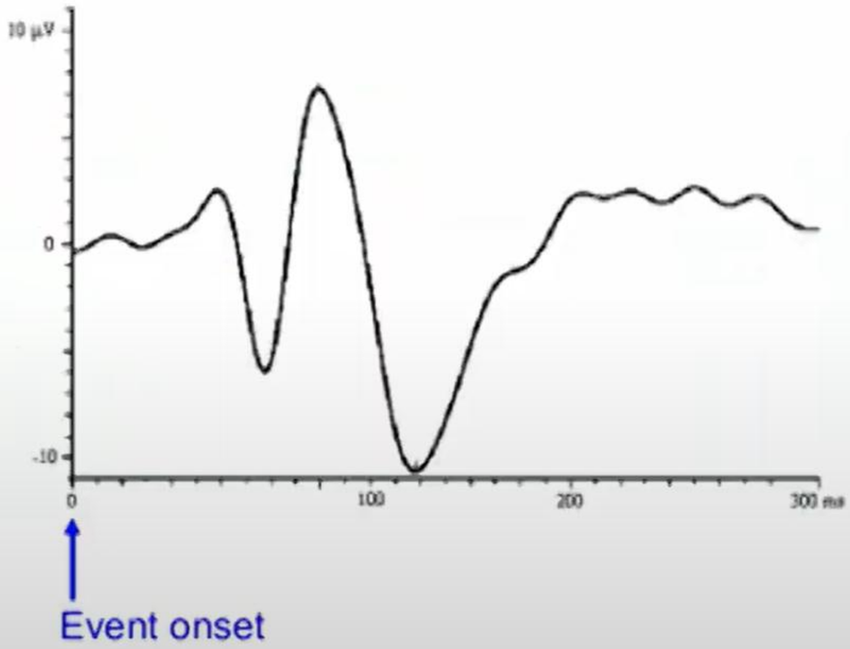
From EEG to event-related potentials (ERPs)

EEG
(electroencephalogram)

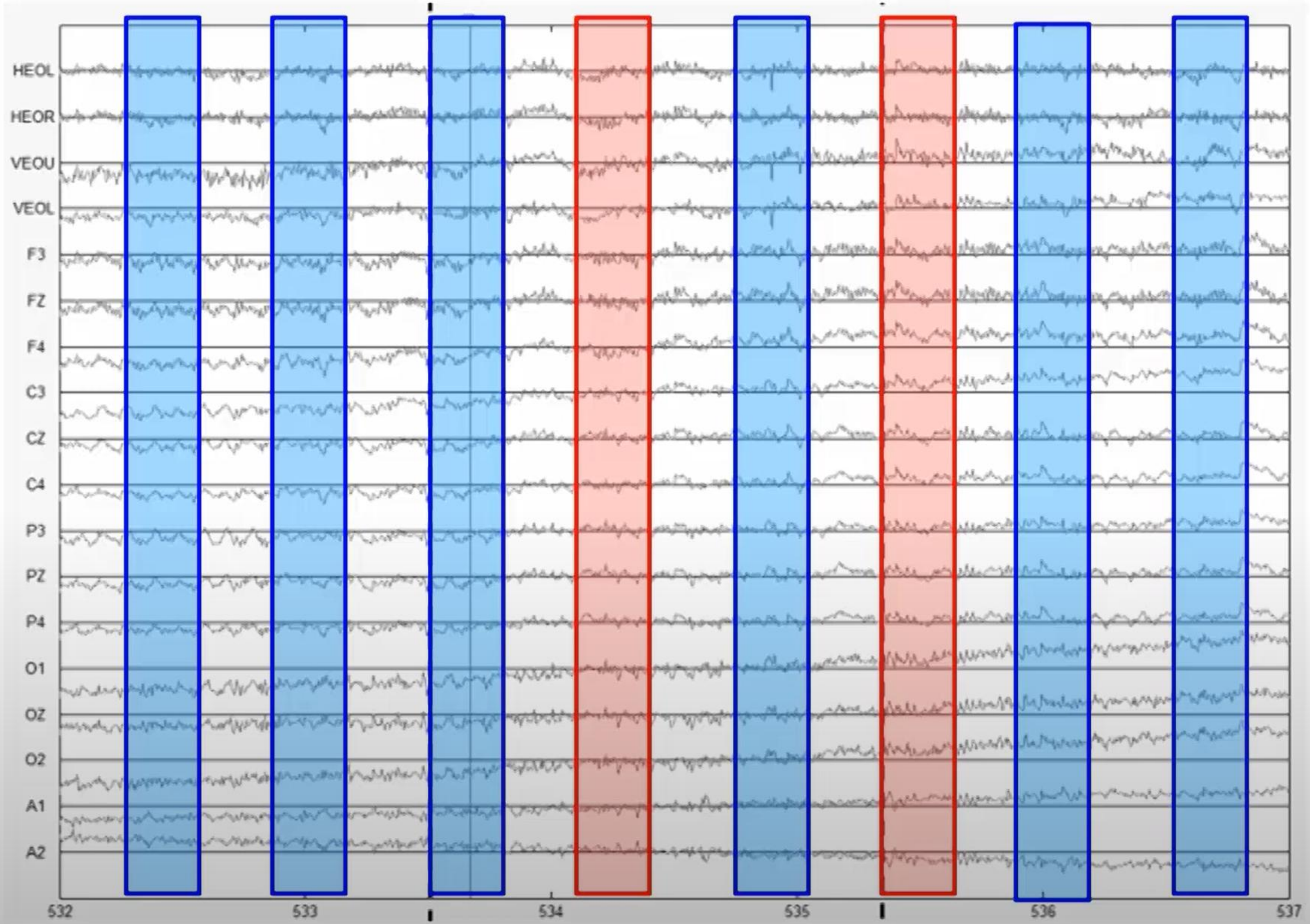


http://www.electropsychology.com/valovi.gif

ERPs
(event-related potentials)

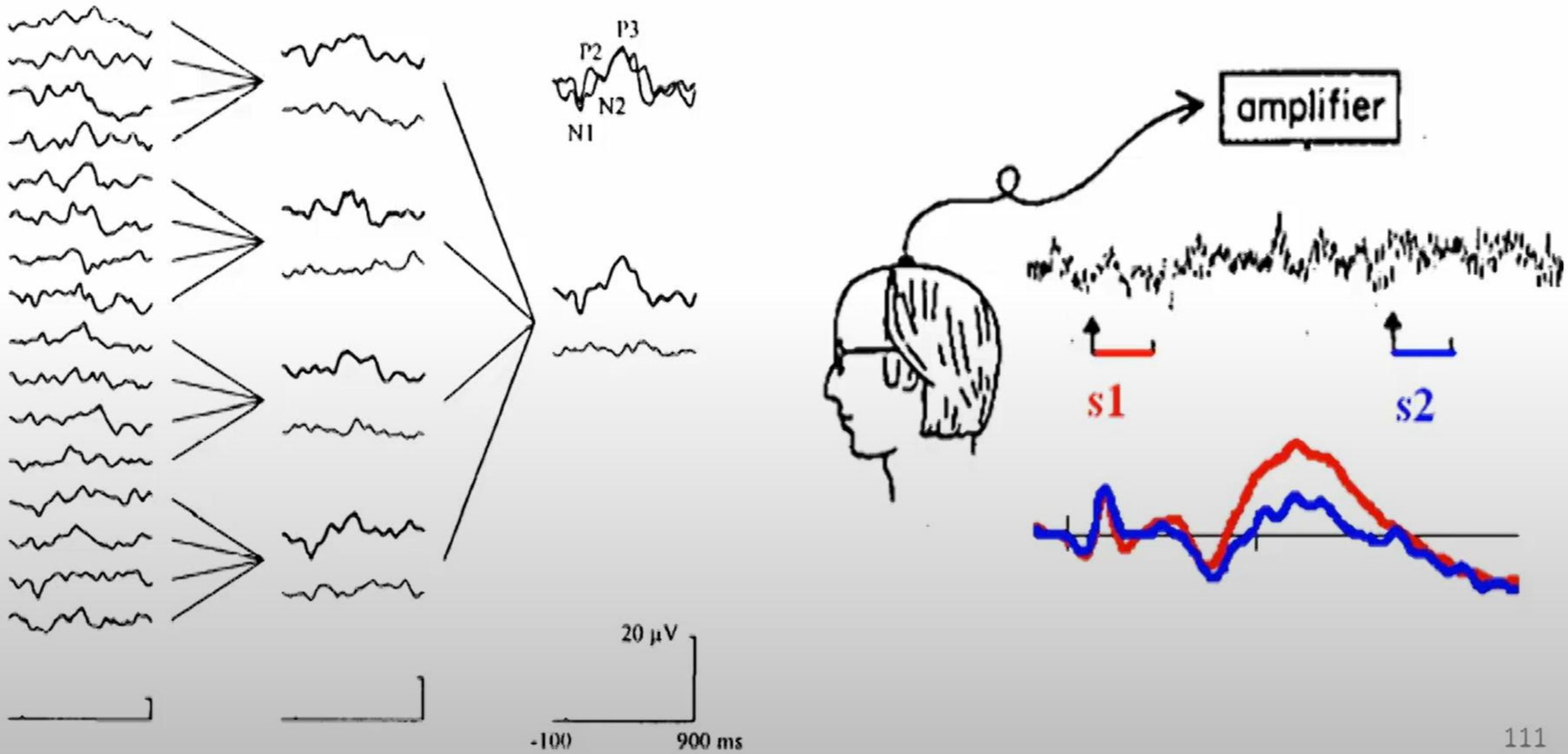


EEG Epochs time-locked to events



Averaging EEG epochs to ERPs

- ERPs are extracted from epochs of EEG associated with stimuli of the same category, and reflect the neural processing of experimental stimuli.

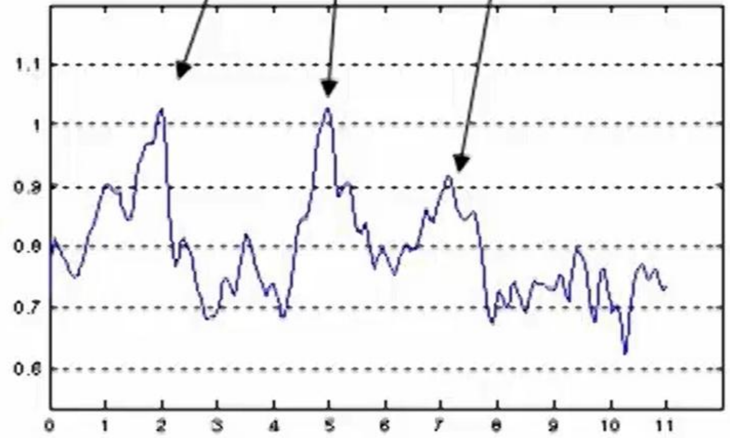


Peak I. - distal cochlear nerve

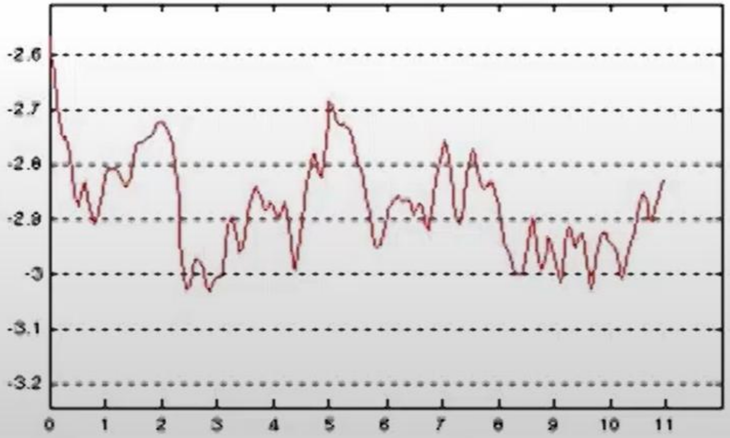
Peak III. - cochlear nucleus

Peak V. - lateral lemniscus

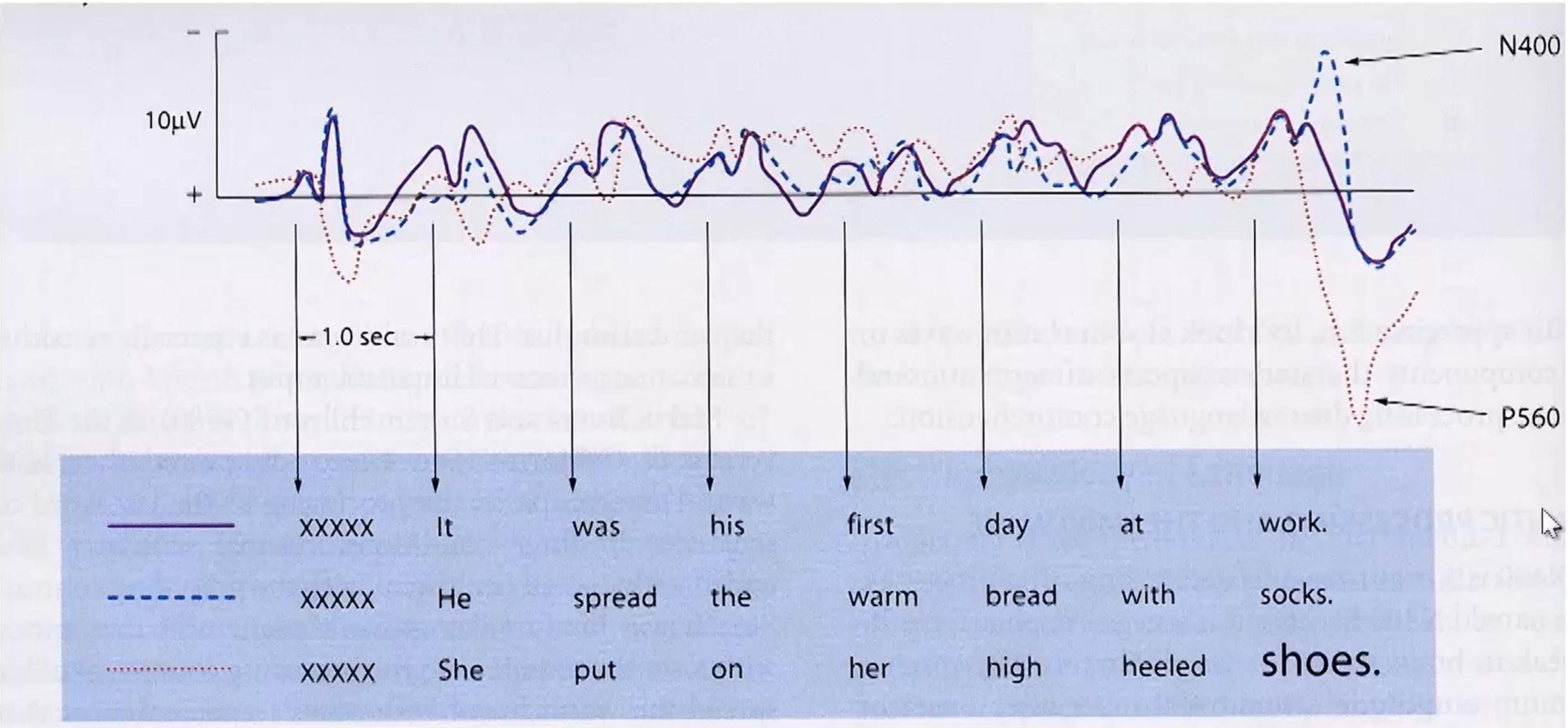
Left ear



Right ear



N400: Semantic Violation detector



Semantically inappropriate words elicited a larger N400 amplitude than semantically appropriate words.